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CHC Helikopter Service

Emergency Checklist S92A

Valid for AMS 8.0 and 9.0

- 1. MAINTAIN AIRCRAFT CONTROL**
- 2. ANALYSE THE SITUATION**
- 3. TAKE APPROPRIATE ACTION**

Publication date: 01 Oct 2024

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Summary of changes

Unit	Page	Remarks
FM	FM-i	Added colour to index and header to match corresponding warning / caution
FM	FM-iii	'APU fuel consumption' updated content
FIRE	FIRE-9	1.5 'APU FIRE IN FLIGHT' added list item to considerations
FIRE	FIRE-12	1.7 'FIRE / SMOKE IN CABIN OR COCKPIT' added item
ENG	ENG-11	2.7 'INTENTIONAL ENGINE SHUTDOWN IN FLIGHT' updated actions
ENG	ENG-13	2.8 'ENGINE RESTART IN FLIGHT' updated actions and considerations
ENG	ENG-15	2.9 'COMPRESSOR STALL' updated actions
ENG	ENG-16	2.10 'FADEC FAULT or FADEC NO DSPTCH' renamed to 'FADEC FAULT', reorganised drill, and amended with updated information
ENG	ENG-17	2.11 'FADEC NO DSPTCH' added new drill with a part of content from drill 2.10
ENG	ENG-24	2.18 'ENGINE FIXED POWER' updated actions
FUEL	FUEL-3	4.2 'FUEL FILTER BYPASS' added 'and' and updated actions
FUEL	FUEL-7	4.6 'AUX FUEL TRANSFER' renamed to 'STOP AUX FUEL TRANSFER'
FUEL	FUEL-8	4.7 'AUX FUEL TRANSFER FAIL' amended 'quality' to 'quantity'
FUEL	FUEL-10	4.9 'AUX FUEL SYSTEM DISPLAY DEGRADED' added new drill
PWRTN	PWRTN-8	5.6 'MGB OIL PRES – CAUTION' amended with updated information
PWRTN	PWRTN-10	5.7 'MGB OIL PRES – WARNING' amended with updated information
PWRTN	PWRTN-12	5.8 'MGB OIL OUT' added warning
PWRTN	PWRTN-13	5.9 'INPUT / ACC HOT' updated actions
PWRTN	PWRTN-15	5.10 'MGB PUMP FAIL' updated actions
PWRTN	PWRTN-18	5.11 'MGB OIL HOT' added note and updated actions
PWRTN	PWRTN-20	5.12 'MGB CAUTION SYSTEM FAULT' moved warning to the end of a list and updated considerations

PRE	Unit	Page	Remarks
FIRE	PWRTN	PWRTN-21	5.13 'MGB OIL PRESSURE CAUTION SYSTEM FAULT' moved warning to the end of a list
ENG	T/RTR	T/RTR-4	6.3 'TAIL ROTOR LOSS OF HYDRAULICS' updated indications
APU	HYD	HYD-7	7.5 'HYDRAULIC SYSTEM OVER-TEMPERATURE' added 'or' to the table
FUEL	HYD	HYD-14	7.12 'SAS PRESSURE' amended 'or' with 'and / or'
PWRTN	ELEC	ELEC-2	8.1 'AC GENERATOR – SINGLE FAILURE' amended with updated information
	ELEC	ELEC-3	8.2 'AC GENERATOR – DUAL FAILURE' added 'Confirm'
T/RTR	ELEC	ELEC-6	8.3 'AC GENERATOR CONTROL UNIT FAILURE' added 'Confirm'
HYD	ELEC	ELEC-13	8.9 'DC CONVERTER DUAL FAILURE AND BACKUP CONVERTER FAILURE' amended 'warnings' with 'cautions' and added note
ELEC	ELEC	ELEC-17	8.11 'BATTERY OFF' updated indications and considerations
CTRL	CTRL	CTRL-7	9.5 'AHRS – SINGLE FAILURE' updated indications
	CTRL	CTRL-8	9.6 'AHRS – DUAL FAILURE' updated indications
A/ICE	CTRL	CTRL-11	9.8 'AUTOPILOT – SINGLE FAILURE' updated indications
RIPS	CTRL	CTRL-12	9.9 'AUTOPILOT – DUAL FAILURE' updated indications
MISC	A/ICE	A/ICE-2	10.1 'PITOT HEAT FAILURE' updated considerations
HOIST	RIPS	RIPS-7	11.6 'DUAL RIPS LWC SENSOR FAILURE' amended 'DGRD' to 'DEGRD'
LAND	RIPS	RIPS-10	11.9 'DUAL RIPS OAT SENSOR FAILURE' amended 'DGRD' to 'DEGRD'
SUPP	MISC	MISC-3	12.1 'ADC – SINGLE FAILURE' amended with updated information
	MISC	MISC-6	12.4 'AVC SYSTEM MALFUNCTION' updated actions
CAS	MISC	MISC-13	12.10 'DCU – SINGLE FAILURE OR MISCOMPARE' amended with updated information
	MISC	MISC-16	12.11 'DCU – DUAL FAILURE' added 'or' to the indications

PRE	Unit	Page	Remarks
FIRE	MISC	MISC-18	12.13 'ELT UNINTENTIONAL OPERATION' renamed to 'CPI/AD ELT UNINTENTIONAL OPERATION' and updated indications
ENG	MISC	MISC-22	12.17 'MAINTENANCE COMPUTER FAILURE' updated considerations
APU	HOIST	HOIST-2	13.1 'HOIST OVERTEMP' amended with updated information
FUEL	HOIST	HOIST-3	13.2 'HOIST CONTROL FAILURE' amended with updated information
PWRTN	HOIST	HOIST-5	13.3 'HOIST CONTROL RUNAWAY' amended with updated information
T/RTR	HOIST	HOIST-7	13.5 'HOIST EMERGENCY CHANGEOVER WITH THIRD PARTY ON HOIST' renamed to 'HOIST EMERGENCY RECOVERY OF THIRD PARTY ON HOIST' and amended in its majority
HYD	LAND	LAND-4	14.3 'PILOT INCAPACITATION' amended content to standardise between all helicopter types
ELEC	LAND	LAND-5	14.4 'PASSENGER BRIEFINGS' updated list item (c)
CTRL	LAND	LAND-6	14.5 'DITCHING / EMERGENCY LANDING' updated actions
A/ICE	SUPP	SUPP-2	SUPP. 1 'CB PANELS' added new section
RIPS	SUPP	SUPP-5	SUPP. 2 'CB LAYOUT' added new section
MISC	SUPP	SUPP-6	SUPP. 3 'AC PRIMARY BUS 1 AND 2' added new section
HOIST	SUPP	SUPP-7	SUPP. 4 'DC PRIMARY BUS 1 AND 2' added new section
LAND	SUPP	SUPP-10	SUPP. 5 'DC ESS AND BATT BUS' added new section
SUPP	SUPP	SUPP-13	SUPP. 6 'BATT HOLDUP AND BATT UTILITY BUS' added new section
CAS	SUPP	SUPP-14	SUPP. 7 'ELECTRICAL DIAGRAM' added new section
	SUPP	SUPP-15	SUPP. 8 'WARNING AND CAUTION LIGHTS' amended 'DGRD' to 'DEGRD'

Preamble

1 Fly the aircraft and maintain N_R

In all emergency / abnormal situations, it is essential that a complete division of responsibility exists, such that PF continues to ensure the safe operation of the aircraft and PM deals with the situation.

The immediate and subsequent action of the PF is to FLY THE AIRCRAFT. PM shall assist PF by monitoring the flight path and the engine and systems instruments.

1. MAINTAIN AIRCRAFT CONTROL
2. ANALYSE THE SITUATION
3. TAKE APPROPRIATE ACTION

Before movement of any throttle, fuel control, or critical systems switch, PM will call for confirmation of selection by PF.

Once the aircraft is under control and on the command of PF, PM shall identify and state all the displayed warnings and cautions, identify the malfunction together with PF, then cancel the **MASTER CAUTION**. It is implicit in the drills that PF will continue to control the aircraft.

Once the immediate actions are complete, if the commander is PF, they should consider handing over the control to the co-pilot in order to better manage the situation.

For all system failures, the drills presume that circuit breakers have been checked. Circuit breakers should not normally be reset. They may be reset once only, provided there are no indications of an electrical fire and the affected system is necessary for the safe recovery of the aircraft.

Drills are preceded by the drill number and title, thus:

01

DRILL TITLE

Indications:

- Caution and warning lights, sounds, faults, and other indications that may occur. To facilitate reading at night, captions that, in the aircraft and in the RFM, appear as red text on a black background are shown as white text on a red background (e.g. **MGB OIL PRES**).

Actions:

1. Immediate actions, outlined in black, shall be carried out from memory, and verified from the checklist

2 Definitions

LAND / DITCH IMMEDIATELY

The consequences of continued flight are likely to be more hazardous than, for example, ditching, or landing in trees. Where the instruction 'Land or ditch immediately' is given in a checklist, the procedure for 'emergency landing' should be followed.

LAND AS SOON AS POSSIBLE

Land at the nearest safe place. Depending on the malfunction and where it occurs, this may require flight to a runway or to the nearest landfall or helideck. The flight should be at an altitude and airspeed that will allow for an immediate landing or ditching if the abnormal condition deteriorates.

LAND AS SOON AS PRACTICAL

Land at the nearest airfield at which technical support can be achieved. If this is not practical, land at a safe landing site chosen for subsequent convenience. The duration of the flight and landing site are at the discretion of the pilot. Extended flight is not recommended.

SAFE OEI FLIGHT CONDITIONS

Safe OEI flight is defined as a sustainable airspeed of not less than V_{TOSS} , the ability to obtain a positive rate of climb at acceptable power levels, and an altitude which provides sufficient clearance from the ground / obstacles so that required manoeuvring can be reasonably achieved. At crew discretion, procedural checks / actions may be carried out while these conditions are being established.

ESTABLISH SAFE FLIGHT CONDITIONS

Establish a safe flight path, while continuing with the emergency procedure, to reduce the time necessary for conduct of a possible emergency landing.

CONTINUE FLIGHT

If the fault persists, or returns after clearing, the decision to continue as planned or to take alternative action shall consider the implications of further deterioration of the affected system.

If the fault clears, the flight may continue as planned without restriction.

WARNING

Draws attention to an operating procedure, practice, or condition that may result in injury or death if not carefully observed or followed.

Where warnings are included between steps within a drill, they are in *italic* (to differentiate them from drill steps) and come before the step to which the warning applies.

CAUTION

Draws attention to an operating procedure, practice, or condition that may damage equipment if not carefully observed or followed.

Where cautions are included between steps within a drill, they are in *italic* (to differentiate them from drill steps) and come before the step to which the caution applies.

NOTE

Draws attention to an operating procedure, practice, or condition that is essential to emphasise.

Where notes are included between steps within a drill, they are in *italic* (to differentiate them from drill steps) and come after the step to which the note applies.

CONSIDERATIONS

Addresses additional information that may be applicable to the checklist or malfunction. This information shall be consulted after completion of the ECL drill when safe to do so.

EMERGENCY TRANSPONDER CODES

Emergency	7700
Communication failure	7600
Unlawful interference (hijack)	7500

3

Instruction cautions not requiring ECL verification

No reference to the ECL is required for 'instruction' cautions. Those drills are marked green.

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APU fuel consumption

APU fuel consumption is 175 lb/hour (80 kg/hour).

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FIRE

1.1	ENGINE FIRE ON THE GROUND
1.2	ENGINE FIRE IN FLIGHT
1.3	POST SHUTDOWN FIRE
1.4	APU FIRE ON THE GROUND
1.5	APU FIRE IN FLIGHT
1.6	SMOKE IN BAGGAGE COMPARTMENT
1.7	FIRE / SMOKE IN CABIN OR COCKPIT

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1.1ENGINE FIRE ON THE GROUND

Indications:



“FIRE ENGINE 1”
or “FIRE ENGINE 2”

Confirming:

Use ATC, other aircraft, or ground crew to confirm fire. If unable to get assistance to confirm, continue as if FIRE IS CONFIRMED.

Caution:

Intermittent fire warnings may be spurious and not indicate a real fire. Continue to monitor. If additional indications appear, take appropriate action.

Actions:

1. Confirm fire:

FIRE IS CONFIRMED	FIRE IS NOT CONFIRMED
Continue with the drill	Go to step 8

2. Throttle (affected engine) STOP
3. FIRE / ARM button (affected engine) Press
4. Fire extinguisher MAIN (forward)
5. If fire is not extinguished approximately 15 seconds after step 4:

a. Fire extinguisher RESERVE (back)
6. If fire is out Perform normal shutdown
- END -----
7. If fire persists, see [14.7 EMERGENCY EVACUATION ON LAND](#)
- END -----
8. If fire is not confirmed visually:

a. Throttle (affected engine) STOP

b. Fuel selector (affected engine) OFF

c. Radio call Complete

----- END -----

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Considerations:

- If fire is not confirmed, consider the engine cool-down period. If engine is shut down without achieving the required cool-down period, see OMB chapter 1 'Limitations'.

1.2ENGINE FIRE IN FLIGHT

Indications:

Primary indications:



“FIRE ENGINE 1”
or “FIRE ENGINE 2”

A steady fire warning light and / or audio warning without secondary indications shall initially be treated as a real fire.

Secondary indications:

- Possibly high TGT
- Smoke and / or flames and / or burning smell

Caution:

Intermittent fire warnings may be spurious and not indicate a real fire. Continue to monitor. If secondary indications appear, take appropriate action.

Note:

In SAR or in HESLO operations, when in or close to the hover, steps 3 to 7 may be actioned as memory items.

Actions:

1. Steady cockpit fire warning	Confirm
2. Safe OEI flight	Attain
3. Safe flight conditions	Establish
4. Throttle (affected engine)	STOP
5. FIRE / ARM button (affected engine)	Press
6. Fire extinguisher	MAIN (forward)

Note:

Pressing the illuminated FIRE / ARM button and deploying the MAIN extinguisher should put out most engine fires.

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7. Radio call As required
8. **If fire is not extinguished approximately 15 seconds after step 6:**
- a. Fire extinguisher RESERVE (back)

9. **If fire is not extinguished after 15 seconds:**
- Steady fire warning light and any of the following secondary indications are confirmed by crew, passengers, other aircraft, nearby installations, or other external sources:
- Possibly high TGT
 - Smoke and / or flames and / or burning smell

Caution:

It is essential that confirmation of a fire by secondary indications is obtained before initiating an immediate ditching or landing.

- a. **LAND / DITCH IMMEDIATELY;** see [14.5 DITCHING / EMERGENCY LANDING](#)

----- END -----

10. **If fire is out or if fire is NOT confirmed by secondary indications:**
- a. Land as soon as possible
- b. Disarm the affected FIRE / ARM button
- c. See [2.7 INTENTIONAL ENGINE SHUTDOWN IN FLIGHT](#)

----- END -----

Considerations:

- Spurious warnings are characterised by incomplete (partial) audio warnings, intermittent flashes of the **FIRE** warning, or by the **FIRE** warning going out after one cycle of the audio warning. If the intermittent warnings continue for a long time, consider [2.7 INTENTIONAL ENGINE SHUTDOWN IN FLIGHT](#).

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1.3

POST SHUTDOWN FIRE

Indications:
Smoke and / or flames in exhaust or TGT >540°C

- Actions:**
1. Throttle (affected engine) STOP

2. ENG START button (affected engine) Press

a. Motor until TGT <250°C
3. If fire persists – PF:
- a. Affected engine Continue motoring
- b. Radio call Complete
4. If fire persists – PM:
- a. Passengers; see [14.7 EMERGENCY EVACUATION ON LAND](#)
- END -----

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1.4

APU FIRE ON THE GROUND

Indications:



🔊 "FIRE APU"

- Note:**
- a. If APU is not running, treat as a false fire warning. Do not start the APU. Connect external power for shutdown.

Confirming:

Use ATC, other aircraft or ground crew to confirm fire. If unable to get assistance to confirm, continue as if FIRE IS CONFIRMED.

Actions:

1. Confirm fire:

FIRE IS CONFIRMED	FIRE IS <u>NOT</u> CONFIRMED
Continue with the drill	a. APU CTRL OFF b. Radio call Complete ----- END -----

- 2. APU CTRL OFF
- 3. APU FIRE / ARM button Press
- 4. Fire extinguisher MAIN (forward)
- 5. PM Prepare for emergency evacuation
- 6. **If fire is not extinguished approximately 15 seconds after step 4:**
 - a. Fire extinguisher RESERVE (back)

7. If:

FIRE IS OUT	FIRE PERSIST
Perform a shutdown (no APU) ----- END -----	See 14.7 EMERGENCY EVACUATION ON LAND ----- END -----

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Considerations:

- The APU and no.2 engine bay are fed from the same fire extinguisher line. If a fire in the no.1 engine bay has occurred but been extinguished by the no.1 MAIN before the APU fire, MAIN shall be selected again after the APU FIRE / ARM button is pressed. Selecting RESERVE will fire the no.2 bottle into the no.1 engine bay.

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1.5

APU FIRE IN FLIGHT

Indications:

Primary indications:

FIRE

PRESS TONE OFF

FIRE # APU

🔊

"FIRE APU"

Secondary indications:

Smoke and / or flames and / or burning smell

- Caution:**
- Intermittent fire warnings are not indications of a real fire. Continue to monitor. When APU IS RUNNING (if additional indications appear), take appropriate action.
- Note:**
- a. If APU is not running, treat as a false fire warning. Do not start the APU unless required for safe flight.
 - b. **FIRE** warning and APU FIRE / ARM button may illuminate momentarily when starting the APU in some flight regimes

Actions:

APU IS RUNNING	APU IS <u>NOT</u> RUNNING
Continue with the drill	Land as soon as practical
	----- END -----

1. Steady cockpit fire warning Confirm
2. Safe flight conditions Establish
3. APU CTRL OFF
4. APU FIRE / ARM button Press
5. Fire extinguisher MAIN (forward)

Note:

Pressing the illuminated FIRE / ARM button and deploying the MAIN extinguisher should put out most APU fires.

6. Radio call Complete
7. If fire is extinguished Go to step 10

Continued on next page

8. If fire is not extinguished approximately 15 seconds after step 5:

- a. Fire extinguisher RESERVE (back)

9. If fire is not extinguished after 15 seconds

Steady fire warning light and any of the following secondary indications are confirmed by crew, passengers, other aircraft, nearby installations, or other external sources:

- Smoke and / or flames and / or burning smell

Caution:

It is essential that confirmation of a fire by secondary indications is obtained before initiating an immediate ditching or landing.

- a. **LAND / DITCH IMMEDIATELY;** see [14.5 DITCHING / EMERGENCY LANDING](#)

----- END -----

10. If fire is out or if fire is NOT confirmed by secondary indications:

- a. Land as soon as possible
- b. Radio call Complete
- c. Passengers Brief
- d. Satellite communication Emergency
- e. APU FIRE / ARM button Disarm / press

----- END -----

Considerations:

- Spurious warnings have been characterised by incomplete (partial) audio warnings, intermittent flashes of the **FIRE** warning, or by the **FIRE** warning going out after one cycle of the audio warning
- Use external power for shutdown
- The APU and no.2 engine bay are fed from the same fire extinguisher line. If a fire in the no.1 engine bay has occurred but been extinguished by the no.1 MAIN before the APU fire, MAIN shall be selected again after the APU FIRE / ARM button is pressed. Selecting RESERVE will fire the no.2 bottle into the no.1 engine bay.

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1.6

SMOKE IN BAGGAGE COMPARTMENT

Indications:

Primary indications:



🔊 “SMOKE IN BAGGAGE”

Secondary indications:

Smoke and / or flames and / or burning smell

Actions:

1. ECS (both cabin and cockpit) VENT
2. Land as soon as possible
3. Radio call Complete
4. Passengers Brief
5. Satellite communication Emergency
6. If fire is confirmed with secondary indications, and fire / smoke persists:

a. **LAND / DITCH IMMEDIATELY**; see [14.5 DITCHING / EMERGENCY LANDING](#)

----- END -----

Considerations:

- If smoke from a baggage compartment fire is observed in the cabin, decreasing airspeed below 140 KIAS may slow or stop the smoke intrusion
- Prior to slowing the aircraft, the commander shall weigh the necessity for speed to quickly reach the nearest safe landing site versus the amount of smoke in the cabin

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1.7

FIRE / SMOKE IN CABIN OR COCKPIT

Indications:

Smoke / fumes / fire or possibly smell of electrical burning

A. ON THE GROUND:

Actions:

1. PF: Radio call

Complete
2. PM: Passengers

Brief
3. Both throttles

STOP

4. Rotor brake Apply
5. All electrical switches OFF
6. Passengers evacuate; see [14.7 EMERGENCY EVACUATION ON LAND](#)

----- END -----

B. IN FLIGHT, SMOKE IS NOT ELECTRICAL:

Actions:

1. Couple 3-cue
2. Both cockpit side windows Open
3. ECS (both cabin and cockpit) Vent (unless smoke is coming from the vents)
- a. If cabin is not occupied, cabin ECS OFF
4. Land as soon as possible
5. Radio call Complete
6. Passengers Brief
7. Satellite communication Emergency
8. PM Fight fire with portable extinguisher
9. If fire / smoke persists:
- a. LAND / DITCH IMMEDIATELY; see [14.5 DITCHING / EMERGENCY LANDING](#)

----- END -----

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C. IN FLIGHT, SMOKE IS OF ELECTRICAL ORIGIN:

Caution:

Prior to shutting off electrical power, the commander shall consider what equipment is essential for continued safe flight.

Actions:

- Affected equipment OFF
- Affected panel lighting OFF
- Attain VMC if possible, without descending below safe altitude
- ECS (cabin and cockpit) Vent (unless smoke is coming from the vents)
- Land as soon as possible
- Radio call Complete
- Passengers Brief
- Satellite communication Emergency
- Utility light As required
- Landing gear Down
- If fire / smoke persists:

Caution:

The left hand and centre MFDs will be lost as well as the flight controls will be reduced to RHS trim in pitch only. Loss of trim on LHS and in roll for RHS. SAS mode in roll despite ATT indicated on AFCS panel – treat as if in SAS mode.

- DC CONV 1, DC CONV 2 OFF
- See [8.8 DC CONVERTER – DUAL FAILURE](#) considerations.

12. If fire / smoke still persists:

Caution:

AHRS input will be lost, prepare for flight on STBY instruments.

- Select COMP mode on PFD
- AC GEN 1, AC GEN 2, APU GEN OFF

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13. If fire / smoke still persists:

Warning:

- a. All MFDs will be lost
- b. BATT must be selected ON again to arm and deploy the floats
- c. ICS will be lost

- a. BATT OFF

14. If fire / smoke still persists:

- a. **LAND / DITCH IMMEDIATELY;** see [14.5 DITCHING / EMERGENCY LANDING](#)

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2.1	ENGINE – DUAL FAILURE
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2.3	ENGINE OVERSPEED
2.4	ENGINE HOT START
2.5	ENGINE OIL SYSTEM FAULTS
2.6	ENGINE CHIP
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2.14	FADEC CHANNEL – DUAL FAILURE
2.15	FADEC CROSS ENGINE COMM FAULT
2.16	ENGINE POWER LIMITING
2.17	ENGINE RATE LIMITING
2.18	ENGINE FIXED POWER
2.19	LIMIT EXCEEDED
2.20	ENGINE OVERSPEED TEST
2.21	STARTER
2.22	TRAINING DISABLED

2.1

ENGINE – DUAL FAILURE

Indications:

- 1 ENG OUT, 2 ENG OUT, and LO ROTOR RPM
- Red X over both engine pods
- Aircraft nose yaws left
-  “ENGINE ONE FAILURE” and “ENGINE TWO FAILURE” and “LOW ROTOR”
- Decrease in all engine indications

Warning:

N_R will decay rapidly even if recovery action is taken within 1 second. Maintain full down collective until N_R recovers.

Actions:

1. Autorotation	Enter immediately (80 to 100 KIAS, 105% N _R)
2. APU CTRL	ON
3. Landing gear	Down
4. Radio call	Complete
5. If over water:	
a. Floats	ARM

Note:

Once the APU is ON, an engine restart will require approximately 45 seconds and at least 2000 feet at minimum ROD speed of 80 KIAS until power is available. Only attempt a restart if time and altitude permit.

Continued on next page

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6. ENG 1 or 2 Attempt emergency restart
- a. Fuel DCT
- b. Throttle STOP
- c. ENG Start button Press as soon as APU is online

Note:

Starter will not engage until N_G is 15% or less.

- d. Throttle (affected engine) FLY

Caution:

During normal engine start procedures, hot-start prevention is automatically provided by the FADEC. This protective feature is not present during an emergency start procedure. The pilot shall monitor engine temperature and manually abort the start by placing the throttle to STOP if a hot start appears imminent.

7. Passengers Brief
8. Satellite communication Emergency

9. If relight unsuccessful:

- a. Throttles STOP
- b. Fuel selectors OFF

----- END -----

Considerations:

- For minimum ROD in autorotation, fly at 80 KIAS and for best range, fly at 100 KIAS. Adjust to 85 KIAS by 300 feet AGL.
- Flare at 150 feet, vary the nose-up attitude according to the landing surface. Reduce to 10° nose-up by 20 feet above the surface and use all available collective to cushion the touchdown.
- As the collective is raised, the collective to yaw interlink will add tail rotor pitch (causing left yaw). Correct with right pedal and control roll attitude with cyclic. As N_R droops below 96%, the AC generators will drop offline at touchdown and the LH pilot's MFDs will go blank unless APU is on and the APU generator switch is on.
- Immediately after landing decrease collective, neutralise cyclic and apply the brakes

Indications:

- **1 ENG OUT** or **2 ENG OUT**
- Red **X** over appropriate engine pod
-  "ENGINE ONE FAILURE"
or "ENGINE TWO FAILURE"

- Possible **ENG 1 OVERSPEED** or **ENG 2 OVERSPEED**

Caution:

The FADEC should immediately recognise a high-speed shaft failure and automatically shut down the engine. In the event that an automatic shutdown does not occur, quick pilot action is required to shut down the affected engine before the severed shaft can damage other aircraft components. The Engine Overspeed caution **may** extinguish after the automatic engine shutdown.

- Torque (affected engine) decreases to zero
- N_G (affected engine) below 55%
- N_P (affected engine) decreases

Warning:

Do not select 2-minute power until obstacle clearance is assured.

Caution:

- If the engine has shut down due to a dual FADEC channel failure (**FADEC A FAIL** and **FADEC B FAIL** illuminated for the same engine):
 - There will be no **ENG OUT** warning, no audio "Engine failure" warning, and no red **X** over the failed pod
 - Collective coupling will not be available because the AMS has lost TGT input even though the remaining engine is operating normally

Continued on next page

Actions:

1. Safe OEI flight Attain
2. Operating engine Select MCP when 5 seconds or more OEI 2-minute power still available
3. APU CTRL ON
4. APU GEN Confirm ON
5. RIPS (in icing conditions) Manual HEAVY mode
6. TCAS II [I] TA ONLY
7. Engine restart Evaluate
 - a. If restart attempted, see [2.8 ENGINE RESTART IN FLIGHT](#)
 - b. If engine to be shut down, see [2.7 INTENTIONAL ENGINE SHUTDOWN IN FLIGHT](#)
- END -----

Considerations:

- See [section 4 APU fuel consumption](#)

2.3ENGINE OVERSPEED

Indications:

- **ENG 1 OVERSPEED** or **ENG 2 OVERSPEED**, and
- N_R and N_P of affected engine cycle between 118% and 120%, or
- N_G of affected engine cycles above 108.5%

Actions:

1. Safe OEI flight	Attain
2. Throttle (affected engine)	IDLE
3. N_R and good engine N_P	Confirm normal
4. Throttle (affected engine)	STOP

5. Affected engine	Shutdown, see 2.7 INTENTIONAL ENGINE SHUTDOWN IN FLIGHT
--------------------------	---

----- END -----

Note:

N_P above 120% or N_G above 108.5% are overspeed conditions, which will cause the FADEC to shut off fuel to the affected engine. The FADEC will automatically restart the engine once N_G and N_P are within limits.

Considerations:

- If torque 0% on affected engine, see [2.2 ENGINE – SINGLE FAILURE](#)
- If the high-speed shaft breaks, auto-reignition will be inhibited and the engine will shut down

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2.4ENGINE HOT START

Indications:

- ENG 1 HOT START or ENG 2 HOT START
- TGT >1000°C

Note:

The FADEC hot-start prevention logic should automatically abort a normal start attempt when TGT reaches 1000°C. Automatic hot-start prevention is not provided during emergency start procedures (throttle forward of IDLE).

Actions:

- Throttle (affected engine) STOP
- Affected engine starter button Press
- Motor engine until TGT <250°C
- Do not attempt to restart engine until cleared by maintenance

----- END -----

Considerations:

- The caution will reset when the affected throttle is moved to OFF

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2.5

ENGINE OIL SYSTEM FAULTS

Indications:

- **ENG 1 OIL PRES** or **ENG 2 OIL PRES**, or
- **ENG 1 OIL HOT** or **ENG 2 OIL HOT**, or
- **ENG 1 OIL BYPASS** or **ENG 2 OIL BYPASS**

Caution:

Do not shut down an engine for an oil system fault if the engine is required for safe flight.

Actions:

1. Safe OEI flight Attain
2. Affected engine Confirm indications
3. Throttle (affected engine) IDLE
4. APU CTRL ON
5. APU GEN Confirm ON
6. TCAS II [I] TA ONLY
7. Land as soon as practical
8. **If oil temperature rising and pressure decreasing:**
 - a. Affected engine Shut down; see [2.7 INTENTIONAL ENGINE SHUTDOWN IN FLIGHT](#)

----- END -----

9. **If engine was not shut down and dual-engine power required for landing:**
 - a. Throttle (affected engine) FLY on final

Caution:

If the throttle is left at IDLE, OEI 30S will not arm automatically.

- b. Affected engine Shut down after touchdown

----- END -----

Considerations:

- **ENG 1/2 OIL BYPASS** may come on regardless of OAT because of the relatively high oil viscosity on start and possibly depending on the degree of contamination in the oil filter. When the engine oil temperature reaches approximately 38°C after start-up, the caution should go out. Set throttles to 50 to 54% N_R until caution is out. If the caution remains on with engine oil temperature above 45°C, shut down and report to maintenance.
- See [section 4 APU fuel consumption](#)

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2.6ENGINE CHIP

Indications:

- ENG 1 CHIP or ENG 2 CHIP

Note:

Indicates metal particles in the affected oil system. No action is required until safe single-engine airspeed is obtained.

Actions:

- Safe OEI flight Attain
- Throttle (affected engine) IDLE
- APU CTRL ON
- APU GEN Confirm ON
- TCAS II [I] TA ONLY
- Land as soon as practical
- If dual-engine power is required for landing:**
 - Throttle (affected engine) FLY on final approach

Caution:

If the throttle is left at IDLE, OEI 30S will not arm automatically.

- Affected engine Shut down after touchdown
- If abnormal noises or signs of engine distress occur:**
 - Affected engine Shut down; see [2.7 INTENTIONAL ENGINE SHUTDOWN IN FLIGHT](#)

----- END -----

Considerations:

- See [section 4 APU fuel consumption](#)

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2.7

INTENTIONAL ENGINE SHUTDOWN IN FLIGHT

Indications:

- Initial cause of failure has been identified and / or an attempted restart has been unsuccessful

Actions:

- Satellite communication Emergency
- TCAS II [I] TA ONLY
- RIPS (when clear of icing) OFF
 - If in icing conditions Select appropriate mode
- Fuel crossfeed As required
- Safe OEI flight Maintain
- APU CTRL ON
- APU GEN Confirm ON
- Throttle (affected engine) STOP
- Fuel selector (affected engine) OFF
- Land as soon as possible
- If TGT remains above 540°C or there is other evidence of combustion after engine shutdown:**
 - ENG START button Press with throttle in STOP
 - TGT below 150°C, ENG START button Press to turn starter off
- Radio call Complete
- Passengers Brief
- RIPS (in icing conditions) Select appropriate mode

Note:

Be aware of performance penalties.

- ECS As required
- FIRE / ARM button Disarm / press

----- END -----

Continued on next page

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Considerations:

- Fuel crossfeed should only be used if required to reach destination or to maintain lateral C of G within limits. OEI consumption at MCP is approximately 1050 lb/hour (475 kg/hour). Alternating between XFEED and DIRECT should be kept to a minimum.
- If there are any erratic engine indications after a fuel switch selection, go back to the previous selection immediately
- If the FMS FUEL page 5/5 shows FAIL for the shut down engine press the LSK to position cursor over FAIL indication, press BACK and then ENTER to reset the readout
- See [section 4 APU fuel consumption](#)

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2.8ENGINE RESTART IN FLIGHT

Warning:

Before attempting a restart, you must be satisfied the engine is undamaged. Inlet anti-ice will have been off while the engine was shut down. Consider potential inlet ice accumulation before restarting a failed engine.

Caution: 1

Use the APU if possible. If doing a cross-bleed start, maintain minimum 90% N_G on the operating engine. Use of engine cross-bleed will result in a loss of up to 3% power of the operating engine.

Note:

In emergency, the throttle of the affected engine may be set directly to FLY after pressing the starter (step 8) although the starter will not actually engage until the N_G is 15% or less. An emergency start is only to be used when the situation dictates that damage to the engine is secondary to the safe continuation of the flight.

Caution: 2

In this case, hot-start protection and automatic blowdown are disabled. Returning the throttle to IDLE after light off will restore hot-start protection, but if a hot-start appears likely, abort by placing the throttle to STOP.

Actions:		
1.	FIRE / ARM button (affected engine)	Disarmed
2.	Heater	OFF
3.	APU CTRL (if time permits)	ON
4.	Throttle (affected engine)	STOP
5.	Fuel selector (affected engine)	DIRECT or XFD
6.	ENG IGN	Auto
7.	AIR SRCE HEAT / START	ENG (if APU is OFF), or APU (if APU is ON)
8.	ENG START button	Press
9.	Throttle (TGT below 150°C)	IDLE
10.	Throttle	FLY
11.	AIR SRCE HEAT / START	ENG

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- 12.TCAS II [I] TA / RA
13.If restart is unsuccessful, see [2.7 INTENTIONAL ENGINE SHUTDOWN IN FLIGHT](#)
14.If restart is successful and conditions permit APU off

----- END -----

Considerations:

- With the AIR SRCE HEAT / START switch in the ENG position and the APU on, the system will use the engine as the air source if the operating engine is approximately 80% torque or above. If the operating engine is below approximately 80% torque, the system will use the APU as the air source.
- If engine was shut-down without the correct cooling procedure, it may be restarted even after the 5-minute window has expired, using the normal start procedure in the event of an emergency
- See [section 4 APU fuel consumption](#)

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2.9

COMPRESSOR STALL

Indications:

- Bang or popping noise from engines, possible yaw kicks
- Rapid increase in TGT, decrease in torque, fluctuations in N_R and N_G

Actions:

1. Collective Reduce
2. Safe OEI flight Attain
3. **If stall still present:**
 - a. Throttle (affected engine) IDLE
4. **If engine recovers:**
 - a. Throttle (affected engine) FLY
5. **If stall persists:**
 - a. Affected engine Shutdown; see [2.7 INTENTIONAL ENGINE SHUTDOWN IN FLIGHT](#)

----- END -----

2.10

FADEC FAULT

Indications:

- **FADEC 1 FAULT** and / or **FADEC 2 FAULT**

Note:

Indicates a fault or combination of faults in the affected FADEC. These faults alone will not affect the engine operation.

Actions:**1. If fault occurs in flight:**

- Note time and date
- Continue flight
- Correct before next takeoff or refer to MEL where applicable

----- END -----

2. If fault occurs on the ground:

- Go to action no.3

3. On the ground:

- Check HLTH page for possible cause. If OEI selector switch is identified as the cause, try to manoeuvre one or both switches as it may clear the fault.

- If caution clears:** Continue flight

----- END -----

- If caution does not clear:** Continue with d
- Shut down affected engine. If both engines are affected, complete a normal shutdown.
- Pull and reset FADEC CBs for the affected engine(s) FADEC 1A/2A (overhead panel) and FADEC 1B/2B (LH panel), see [section SUPP.1 CB panels](#)
- One engine affected:** Restart engine and confirm no FADEC caution
- Both engines affected:** Perform a normal start and confirm no FADEC cautions
- If caution clears:** Continue flight
- If caution does not clear:** Refer to MEL where applicable

----- END -----

Considerations:

- The cause for any or both **FADEC FAULT** may be displayed on HLTH page
- Shutting down the engine with the fault may correct the problem. If it does not, a complete power-off and restart may do.

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2.11 FADEC NO DSPTCH

Indications:

- FADEC 1 NO DSPTCH or FADEC 2 NO DSPTCH

Note:

Indicates a level of sensor redundancy has been lost.

Actions:

1. If fault occurs in flight:

- Continue flight
- Correct before next takeoff

----- END -----

2. If fault occurs on the ground:

- Correct before flight

----- END -----

2.12

FADEC OVERHEAT

Indications:

- **FADEC 1 OVRHEAT** or **FADEC 2 OVRHEAT**
- Possibly **FADEC A/B FAIL** or **ENGINE 1/2 POWER LIM** or **FADEC 1/2 FAULT**

Caution:

FADEC OVRHEAT indicates that both channels of the affected engine FADEC have overheated and component failure could result. If both channels fail, the engine will shut down. See [2.14 FADEC CHANNEL – DUAL FAILURE](#). If a single channel fails, see [2.13 FADEC CHANNEL – SINGLE FAILURE](#).

Actions:

1. If one **FADEC OVRHEAT** illuminates:
 - a. Land as soon as practical
2. If both **FADEC OVRHEAT** illuminate:
 - a. Land as soon as possible
 - b. Radio call Complete
 - c. Passengers Brief
 - d. Satellite communication Emergency

----- END -----

2.13 FADEC CHANNEL – SINGLE FAILURE

Indications:

- FADEC 1A FAIL or FADEC 1B FAIL or FADEC 2A FAIL or FADEC 2B FAIL
- ENG 1 POWER LIM or ENG 2 POWER LIM
- FADEC 1 FAULT or FADEC 2 FAULT
- Channel A failure:
 - Loss of the affected engine oil temperature and TGT indications
 - ENGINE OIL HOT, FUEL BYPASS, and ENG OIL PRES are disabled
- Channel B failure:
 - Loss of the affected engine oil pressure and TGT indications
 - ENG OIL BYPASS, ENG FUEL PRES, and ENG CHIP are disabled

Note:

Complete loss of a single FADEC channel (A or B) will not degrade engine performance.

Warning:

If the second channel fails, engine will shutdown. See [2.14 FADEC CHANNEL – DUAL FAILURE](#).

Actions:

1. Land as soon as practical
2. Correct fault prior to next takeoff

----- END -----

Considerations:

- If coupled 3-cue, collective coupling will be lost because the AMS has lost TGT input

Indications:

- **FADEC 1A FAIL** and **FADEC 1B FAIL**, or **FADEC 2A FAIL** and **FADEC 2B FAIL**
- **FADEC 1 XENG** or **FADEC 2 XENG**
- **ENG 1 POWER LIM** or **ENG 2 POWER LIM**
- **FADEC 1 FAULT** or **FADEC 2 FAULT**
- Affected engine shuts down automatically
- Loss of all indications for the affected engine with red flags on the gauges
- No **ENG OUT** warning, no red **X** over the affected engine pod on the EICAS, and no **"Engine failure"** audio warning for the affected engine

Caution:

OEI 30S will be armed on the remaining engine (the one with **FADEC 1/2 XENG** displayed) 5 seconds after the caution light has illuminated.

Actions:

1. See [2.2 ENGINE – SINGLE FAILURE](#)

----- END -----

Considerations:

- If coupled 3-cue, collective coupling will be lost because the AMS has lost TGT input

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2.15

FADEC CROSS ENGINE COMM FAULT

Indications:

- FADEC 1 XENG or FADEC 2 XENG

Caution:

OEI 30S will be armed on the engine with caution displayed, 5 seconds after the caution light has illuminated.

Actions:

- Avoid high power settings and rapid collective movements
- TCAS II [\[I\]](#) TA ONLY
- Land as soon as practical

----- END -----

Considerations:

- OEI 30S can be deselected once in use, but will automatically arm again after 30 seconds below MCP

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2.16ENGINE POWER LIMITING

Indications:

- ENG 1 POWER LIM or ENG 2 POWER LIM

Actions:

- Avoid high rates of descent
- Avoid high power settings
- TCAS II [\[I\]](#) TA ONLY
- Land as soon as practical

----- END -----

Considerations:

- Indicates an engine control system malfunction. Full power may not be available (the AISBV is probably stuck open) but engine operation is otherwise normal.

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2.17ENGINE RATE LIMITING

Indications:

- ENG 1 RATE LIM and / or ENG 2 RATE LIM
- Possible transient torque split or N_R droop on power application. Torque matching may be inoperative.
- Possible NR next to the N_R gauge

Note:

ENG RATE LIM indicates an engine control system malfunction. Full power is still available, but engine transient response may be degraded with slow acceleration.

Actions:

1. Make collective inputs gently to minimise transient N_R droop
2. TCAS II [I] TA ONLY
3. Land as soon as practical

----- END -----

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2.18

ENGINE FIXED POWER

Indications:

- **ENG 1 FIXED PWR** or **ENG 2 FIXED PWR**
- Affected engine has not shutdown but has a fixed N_G which does not respond to collective movements

Warning:

In a fail fixed condition, there is no automatic engine overspeed protection.

Caution:

If possible, the pilot should fly the final approach and landing using single-engine landing procedure. Changes in airspeed and altitude may cause the malfunctioning engine to either stall or overspeed.

Actions:

1. Collective Control N_R within limits
2. Collective Confirm fail fixed (small collective inputs)
3. Safe OEI flight Attain
4. TCAS II [\[I\]](#) TA ONLY
5. Land as soon as practical
6. **Before starting approach:**
 - a. Airspeed 80 KIAS
 - b. Reduce collective to establish rate of descent or until N_R increases not above 106%
7. **If conditions do not allow the approach to be flown at 80 KIAS or at the required rate of descent with $N_R \leq 106\%$ and good engine torque just above 0%:**
 - a. Affected engine Shutdown; see [2.7 INTENTIONAL ENGINE SHUTDOWN IN FLIGHT](#)

----- END -----

8. **If conditions allow the approach to be flown at 80 KIAS with N_R within limits:**
 - a. Maintain positive torque on the good engine
 - b. Fly normal / shallow approach and landing to wheels light on the ground

Caution:

Lowering collective fully down will cause N_R to exceed 110.

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9. On the ground before lowering the collective:

- a. Throttle (affected engine) STOP

----- END -----

Considerations:

- In this condition, the engine will not respond to throttle commands at any position other than STOP. If moving the throttle to STOP does not cause the engine to shut down, set the affected engine fuel selector to OFF.
- 500 feet/min ROD will allow for a 3° glideslope at 80 KIAS

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CAS

2.19 LIMIT EXCEEDED

Indications:

- **LIMIT EXCEEDED**

Actions:

1. Land as soon as practical

----- END -----

Note:

The limit exceeded caution illuminates when dual or single engine blow away power is activated exceeding a gearbox torque limit.

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2.20 ENGINE OVERSPEED TEST

Indications:

- **ENG 1 OVSP TEST** or **ENG 2 OVSP TEST**

Actions:

- 1. In flight:**
 - Land as soon as practical

----- END -----
- 2. On the ground:**
 - Correct fault prior to next takeoff

----- END -----

Considerations:

- Any **ENG OVSP TEST** indicates that the FADEC failed a portion of the overspeed system test. Overspeed protection for the affected engine may not be available.

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2.21

STARTER

Indications:

- **STARTER 1 ON** or **STARTER 2 ON**

Actions:

1. TGT and N_G (affected engine) Monitor
2. **If engine does not accelerate:**
 - a. Abort start: Throttle (affected engine) STOP
3. **If engine has achieved normal ground idle:**
 - a. Shut down engine: Throttle (affected engine) STOP
4. **If starter is still engaged:**
 - a. APU CTRL OFF

----- END -----

Considerations:

- The **STARTER ON** caution indicates that the associated starter has been motoring for more than 60 seconds

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2.22

TRAINING DISABLED

Indications:

- **TRAIN DISABLED** when DUAL TRAIN is selected
- FADECs will not go into training mode

Note:

If there are FADEC or engine faults, the FADEC will not allow the training feature to be used. Once training mode is selected, only an actual OEI condition will automatically turn off training. Faults occurring when one engine is in TRAIN IDLE will revert the FADEC to dual-engine training mode.

Actions:

1. OEI TRAIN master switch OFF
2. Both collectives Exercise OEI SEL switch through OEI 2M and OEI 30S positions
3. OEI TRAIN master switch DUAL TRAIN
4. **If fault still exists:**
 - a. Continue flight without using training mode

----- END -----

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APU

3.1	APU FAILURE
3.2	APU FAULT
3.3	APU ACCUM LOW – advisory
3.4	APU OVERLOAD
3.5	APU GENERATOR FAILURE
3.6	APU GENERATOR CONTROL UNIT FAILURE

3.1

APU FAILURE

Indications:

- **APU FAIL** indicates an APU automatic shutdown
- **APU ON** is not illuminated with APU CTRL switch in the ON position

Caution:

After automatic APU shutdown, do not attempt to restart the APU.

Actions:

1. In flight:

- a. Attain VMC as soon as possible
- b. Land as soon as practical
- c. APU CTRL OFF
- d. AIR SRCE HEAT / START switch ENG
- e. After landing Continue with action no.2 below

2. On ground – engines running:

- a. APU CTRL OFF

EXT PWR available:		EXT PWR <u>NOT</u> available:
AC EXT PWR – RESET, then ON	DC EXT PWR – ON	Consider disembarking passengers and see considerations
	Lower ramp and verify RAMP OPEN	
	Guard the collective firmly prior to shutdown since the removal of AC will remove the trim force	
Perform normal or high wind shutdown as required		

----- END -----

Considerations:

- The APU generator will not be available for emergency power or for engine shutdown. If APU generator power is not available during engine shutdown, LH MFDs will blank when generators are switched off. Monitor TGT on pilot inboard EICAS display (note, TGT information will disappear below about 12% N_G). In addition, the collective and cyclic trims will fail (loss of DC essential) when the AC generators go offline.
- Once the aircraft is shut down, it cannot be started again until the APU is serviceable

- PRE
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3.2

APU FAULT

Indications:

- **APU FAULT**

Actions:

1. Unless APU required for safety of flight:
 - a. APU CTRL OFF
 - b. AIR SRCE HEAT / START switch ENG
2. Land as soon as practical

----- END -----

Considerations:

- A subsequent APU start may be attempted after the **APU ACCUM LOW** advisory goes out
- If the APU is ON in flight with **APU FAULT** illuminated, it will shut down automatically on landing

- PRE
- FIRE
- ENG
- APU**
- FUEL
- PWRTN
- T/RTR
- HYD
- ELEC
- CTRL
- A/ICE
- RIPS
- MISC
- HOIST
- LAND
- SUPP
- CAS

3.3
APU ACCUM LOW – advisory

Indications:

- **APU ACCUM LOW** with APU CTRL OFF

Actions:

- In-flight:**
 - AIR SRCE HEAT / START switch ENG
 - Land as soon as practical

----- END -----

- On the ground:**

- Correct before flight

----- END -----

Considerations:

- If the ACC RCG CB has tripped (overhead panel, BATT bus, second row no.7 from left) the **APU ACCUM LOW** advisory will not appear

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CAS

3.4

APU OVERLOAD

Indications:

- **APU OVERLOAD**

Actions:

1. Decrease APU generator electrical load (see considerations)
2. **If caution light extinguishes:**
 - a. Continue flight
3. **If caution light remains illuminated:**
 - a. Attain VMC as soon as possible
 - b. Land as soon as practical

----- END -----

----- END -----

Considerations:

- Following a dual-AC generator failure, the **APU OVERLOAD** will automatically illuminate if pilot windshield heat is selected to OVRD and RH engine inlet anti-ice is elected to ON
- The only actions that will reduce the APU generator load sufficiently to extinguish the caution are switching off the RH inlet anti-ice and the RH windshield heat (in OVRD). Do not load shed these items until flight conditions permit.

- PRE
- FIRE
- ENG
- APU**
- FUEL
- PWRTN
- T/RTR
- HYD
- ELEC
- CTRL
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- RIPS
- MISC
- HOIST
- LAND
- SUPP
- CAS

3.5

APU GENERATOR FAILURE

Indications:

- **APU GEN FAIL**

Actions:

1. APU GEN OFF / RESET then ON
2. **If reset unsuccessful:**
 - a. **In flight**
 - i. Attain VMC as soon as possible
 - ii. Land as soon as practical
 - iii. APU GEN OFF
 - iv. After landing Continue with b below
 - b. **On ground – engines running:**
 - i. APU CTRL OFF

EXT PWR available:		EXT PWR <u>NOT</u> available:
AC EXT PWR – RESET, then ON	DC EXT PWR – ON	Consider disembarking passengers and see considerations
	Lower ramp and verify RAMP OPEN	
	Guard the collective firmly prior to shutdown since the removal of AC will remove the trim force	
Perform normal or high wind shutdown as required		

----- END -----

Considerations:

- The APU generator will not be available for emergency power or for engine shutdown. If APU generator power is not available during engine shutdown, LH MFDs will blank when generators are switched off. Monitor TGT on pilot inboard EICAS display (note, TGT information will disappear below about 12% N_G). In addition, the collective and cyclic trims will fail (loss of DC essential) when the AC generators go offline.

- PRE
- FIRE
- ENG
- APU**
- FUEL
- PWRTN
- T/RTR
- HYD
- ELEC
- CTRL
- A/ICE
- RIPS
- MISC
- HOIST
- LAND
- SUPP
- CAS

3.6

APU GENERATOR CONTROL UNIT FAILURE

Indications:

- **APU GCU FAIL**

Actions:

1. APU GEN OFF / RESET then ON
2. **If reset unsuccessful:**
 - a. **In flight**
 - i. Attain VMC as soon as possible
 - ii. Land as soon as practical
 - iii. APU GEN OFF
 - iv. After landing Continue with b below
 - b. **On ground – engines running:**
 - i. APU CTRL OFF
 - ii. Passengers Disembark

EXT PWR available:		EXT PWR <u>NOT</u> available:
AC EXT PWR – RESET, then ON	DC EXT PWR – ON	Disembark passengers and see considerations
	Lower ramp and verify RAMP OPEN	
	Guard the collective firmly prior to shutdown since the removal of AC will remove the trim force	
Perform normal or high wind shutdown as required		

----- END -----

Considerations:

- The APU generator will not be available for emergency power or for engine shutdown. If APU generator power is not available during engine shutdown, LH MFDs will blank when generators are switched off. Monitor TGT on pilot inboard EICAS display (note, TGT information will disappear below about 12% N_G). In addition, the collective and cyclic trims will fail (loss of DC essential) when the AC generators go offline.

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FUEL

4.1	ENGINE FUEL PRESSURE LOW
4.2	FUEL FILTER BYPASS
4.3	PRIME PUMP FAILURE
4.4	PRIME SHUTOFF VALVE WRONG POSITION
4.5	FUEL LOW LEVEL
4.6	STOP AUX FUEL TRANSFER
4.7	AUX FUEL TRANSFER FAIL
4.8	AUX FUEL TRANSFER DEGRADED
4.9	AUX FUEL SYSTEM DISPLAY DEGRADED

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4.1ENGINE FUEL PRESSURE LOW

Indications:

- ENG 1 FUEL PRES or ENG 2 FUEL PRES
- Possibly erratic fuel flow and / or fluctuating engine indications

Actions:

- Fuel selector (affected engine) XFD
- If caution light extinguishes:
 - Land as soon as practical

----- END -----
- If caution light remains illuminated:
 - Safe OEI flight Attain
 - Fuel selector (affected engine) DIRECT
 - Land as soon as practical
 - Be prepared for engine flameout

----- END -----

Considerations:

- This caution indicates reduced fuel pressure at the engine driven fuel boost pump. This could be an indication of an air leak in the fuel supply lines.

4.2

FUEL FILTER BYPASS

Indications:

- FUEL 1 BYPASS and / or FUEL 2 BYPASS

Actions:

1. If only one caution light illuminates:

- a. Fuel selector (affected engine) XFD
- b. Land as soon as practical

----- END -----

2. If both caution lights illuminate:

- a. Both fuel selectors DIRECT
- b. Land as soon as possible
- c. Safe OEI flight Attain
- d. Radio call Complete
- e. Passengers Brief
- f. Satellite communication Emergency

----- END -----

Considerations:

- This is an impending bypass indication

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4.3

PRIME PUMP FAILURE

Indications:

- PRIME PUMP FAIL

Note:

Only one of the two prime pumps will operate at one time. When in AUTO mode, the active pump will switch with each MDC power cycle. The caution indicates that the active prime pump has failed if selected to MANUAL, or that both prime pumps have failed if selected to AUTO.

Actions:

1. If MANUAL selected:

- a. Fuel prime Select alternate pump
- b. Continue flight
- c. Correct before next flight

----- END -----

2. If AUTO selected:

- a. Correct before next flight

----- END -----

Caution:

APU will not be available – see shutdown considerations for APU FAIL ([3.1 APU FAILURE](#)).

Considerations:

- When starting engines with the fuel valves in crossfeed, the PRIME PUMP FAIL caution may illuminate as long as the starter is engaged. This is not an indication of prime pump failure. During an actual prime pump failure, the PRIME PUMP FAIL will remain illuminated after the engine starter cuts out.

PRE

4.4**PRIME SHUTOFF VALVE WRONG POSITION**

FIRE

Indications:

- **SOV 1 WRNG POSN** or **SOV 2 WRNG POSN**

ENG

Actions:

1. Continue flight
2. Correct before next flight

----- END -----

APU

FUEL**Considerations:**

- There are two failure modes of the prime shutoff valve:
 - a. Open when the valve is commanded closed
 - b. Closed when the valve is commanded open
- If the valve fails in the closed position when it should be open, the engine may start very slowly because there will be no prime fuel pressure in the line

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4.5

FUEL LOW LEVEL

Indications:

- **FUEL 1 LOW** or **FUEL 2 LOW**
- Affected side fuel level yellow on EICAS, quantity <215 ±45 lb (98 ±20 kg)

Caution:

When **FUEL LOW** caution(s) are illuminated, avoid steady state pitch attitudes higher than 10° nose-up and lower than 5° nose-down

Actions:

1. Crossfeed As required
2. Land as soon as practical
3. **If both cautions lights illuminate:**
 - a. Land as soon as possible
 - b. Radio call Complete
 - c. Passengers Brief
 - d. Satellite communication Emergency
 - e. TCAS II [I] TA ONLY
 - f. Be prepared for engine flameout

----- END -----

Considerations:

- Fly approach to a running landing to avoid pitch of more than 10° nose-up
- Fuel remaining in the affected tank (approximately 215 lb / 98 kg) is sufficient for no more than 20 minutes running for that engine at cruise power

4.6STOP AUX FUEL TRANSFER

Indications:

- STOP AUX XFER

Actions:

1. IAFS MODE SELECT OFF

----- END -----

Warning:

Failure to initiate manual shutdown of the auxiliary fuel transfer, within 30 seconds of the STOP AUX XFER illuminating, may result in structural damage and rupture of a main fuel tank due to overpressure.

Considerations:

- If aux tank fuel is required for the flight, use the MANUAL transfer mode and monitor the transfer process to prevent filling the main tanks above 2300 lb (1045 kg) each

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4.7

AUX FUEL TRANSFER FAIL

Indications:

- AUX XFER FAIL**
- Main fuel tank levels not increasing

Actions:

- IAFS MODE SELECT OFF
- LH REFUEL and RH REFUEL Confirm OPEN
- IAFS MODE SELECT AUTO
- If main fuel tank quantity increases:**
 - Continue flight

----- END -----

- If main fuel tank quantity does not increase within 60 seconds:**
 - IAFS MODE SELECT MANUAL
- If main fuel tank quantity does not increase within further 60 seconds:**
 - IAFS MODE SELECT OFF

----- END -----

- Continue flight, replan if required (only the fuel in the main tanks is available – update the FMS fuel quantity to indicate the main tanks quantity only)

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4.8 AUX FUEL TRANSFER DEGRADED

Indications:

- **AUX XFER DGRD**
- Yellow fuel flow lines replace the green lines on the EICAS page from the aux fuel tanks to the main fuel tanks
- Auxiliary fuel transfer is slower than normal

Actions:

1. Monitor fuel transfer
2. Continue flight

----- END -----

Considerations:

- Indicate that the fuel transfer rate is slower than normal. This may be due to a single pump failure.

PRE

4.9

AUX FUEL SYSTEM DISPLAY DEGRADED

FIRE

ENG

APU

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Indications:

- Red AUX-A advisory next to the fuel quantity gauge on the EICAS page
- IAFS fuel quantity or status is unknown

Actions:

1. Select IAFS mode select switch to OFF and back to AUTO or MANUAL to reset system
2. Carefully monitor main tank fuel contents to reveal if fuel is transferring

Considerations:

- It does not indicate a failure of the IAFS to transfer fuel to the main tanks

----- END -----

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PWRTN

5.1	SWASHPLATE TEMPERATURE
5.2	SWASHPLATE TEMPERATURE WARNING
5.3	IGB / TGB OIL LEVEL LOW
5.4	IGB / TGB OIL HOT / CHIP
5.5	GEARBOX CHIP
5.6	MGB OIL PRES – CAUTION
5.7	MGB OIL PRES – WARNING
5.8	MGB OIL OUT
5.9	INPUT / ACC HOT
5.10	MGB PUMP FAIL
5.11	MGB OIL HOT
5.12	MGB CAUTION SYSTEM FAULT
5.13	MGB OIL PRESSURE CAUTION SYSTEM FAULT

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5.14	BYPASS TEST FAIL
5.15	MGB MAN COOL
5.16	MGB OIL BYPASS
5.17	MGB SENSOR FAIL

PRE

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CAS

5.1 SWASHPLATE TEMPERATURE

Indications:

- S/P TEMP DETECT

Actions:

1. Land as soon as practical

----- END -----

- PRE
- FIRE
- ENG
- APU
- FUEL
- PWRTN**
- T/RTR
- HYD
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- CAS

5.2

SWASHPLATE TEMPERATURE WARNING

Indications:

- **S/P TEMP LIMIT**
- 🔊 “SWASHPLATE TEMPERATURE”

Actions:

1. Land as soon as possible
2. Radio call Complete
3. Passengers Brief
4. Satellite communication Emergency
5. **If secondary indications such as restricted control movement, unusual noises, or vibrations occur:**
 - a. **LAND / DITCH IMMEDIATELY;** see [14.5 DITCHING / EMERGENCY LANDING](#)

----- END -----

- PRE
- FIRE
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5.3

IGB / TGB OIL LEVEL LOW

Indications:

- **IGB OIL LEVEL** or **TGB OIL LEVEL** prior to engine start

Actions:

1. Oil levels Replenish before start

----- END -----

Considerations:

- The cautions are latched and will not clear on start-up. They can only be cleared by powering off and then topping-up the oil level before powering on again.

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- CAS

5.4 IGB / TGB OIL HOT / CHIP

Indications:

- IGB HOT or TGB HOT or IGB CHIP or TGB CHIP

Actions:

1. Airspeed Reduce to 80 KIAS
2. Land as soon as possible
3. Radio call Complete
4. Passengers Brief
5. Satellite communication Emergency
6. If any unusual loud noises and / or vibrations or pedal kicks develop, or both CHIP and HOT illuminate for one gearbox:
 - a. LAND / DITCH IMMEDIATELY; see [14.5 DITCHING / EMERGENCY LANDING](#)

----- END -----

- PRE
- FIRE
- ENG
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- ELEC
- CTRL
- A/ICE
- RIPS
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- CAS

5.5

GEARBOX CHIP

Indications:

- **MGB CHIP** or **ACC 1 CHIP** or **ACC 2 CHIP** or **INPUT 1 CHIP** or **INPUT 2 CHIP**

Actions:

1. If only one caution light is illuminated:

- a. Land as soon as practical

----- END -----

2. If two or more caution lights are illuminated:

- a. Land as soon as possible
- b. Radio call Complete
- c. Passengers Brief
- d. Satellite communication Emergency

----- END -----

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5.6

MGB OIL PRES – CAUTION

Indications:

- **MGB OIL PRES** with MGB oil pressure below 45 psi but steady above 35 psi

Note:

If **MGB PUMP 1 FAIL** or **MGB PUMP 2 FAIL** is illuminated, see

[5.10 MGB PUMP FAIL](#).

If **MGB OIL BYPASS** is illuminated, see [5.13 MGB OIL PRESSURE CAUTION SYSTEM FAULT](#).

If **INPUT / ACC 1 HOT** or **INPUT / ACC 2 HOT** is illuminated, see

[5.9 INPUT / ACC HOT](#).

Actions:

1. Land as soon as possible
2. Heater OFF
3. APU CTRL ON
4. APU GEN Confirm ON
5. Airspeed 80 to 100 KIAS
6. Radio call Complete
7. Passengers Brief
8. Satellite communication Emergency

Warning:

A total loss of MGB oil pressure may result in a MGB failure within less than 10 minutes.

9. If **ANY** of the following symptoms of an imminent gearbox failure exist:

- **MGB OIL OUT** with MGB oil pressure below 5 psi and
 -  "GEARBOX OIL OUT"
 - Increased power required at constant collective setting and airspeed
 - Yaw kicks
 - Unusual noise or vibrations
 - Loss of two hydraulic pumps
- a. **LAND / DITCH IMMEDIATELY**; see [14.5 DITCHING / EMERGENCY LANDING](#)

----- END -----

Continued on next page

PRE

Note:

If **MGB OIL PRES** is illuminated, see [5.7 MGB OIL PRES – WARNING](#).

FIRE

Caution:

The main AC generators are cooled by MGB oil. Loss of oil or high oil temperatures may result in a generator failure and loss of main electrical power. The APU should be started so the APU generator is available.

ENG

APU

Considerations:

- A loss of main gearbox oil may result in the mechanical failure of a hydraulic pump drive
- See [section 4 APU fuel consumption](#)
- The **MGB OIL PRES** caution illuminates in flight when MGB pressure at the manifold falls below 45 psi or pressure at the last jet falls below 24 psi. The **MGB OIL PRES** warning illuminates when oil pressure at the manifold falls below 35 psi and pressure at the last jet falls below 24 psi. If the **MGB OIL PRES** warning illuminates, the **MGB OIL PRES** caution will not be illuminated.

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5.7

MGB OIL PRES – WARNING

Indications:

- **MGB OIL PRES** with MGB oil pressure below 35 psi and

 “GEARBOX PRESSURE”

- **MGB OIL BYPASS**

Note:

If **MGB PUMP 1 FAIL** or **MGB PUMP 2 FAIL** is illuminated, see [5.10 MGB PUMP FAIL](#).

If **INPUT / ACC 1 HOT** or **INPUT / ACC 2 HOT** is illuminated, see [5.9 INPUT / ACC HOT](#).

Actions:

1. If **MGB OIL PRES** is on but **MGB OIL BYPASS** is not illuminated:
 - a. MGB OIL BYP switch BYP (within 5 sec)

Note:

Avoid rapid or frequent power changes.

2. Descend to a minimum safe altitude while reducing speed to 80 to 100 KIAS
3. Pitch nose up (if practical) Maintain >2°
4. Torque Minimum 15%

Warning:

A total loss of MGB oil pressure may result in MGB failure in less than 10 minutes.

5. If **ANY** of the following symptoms of an imminent gearbox failure exist:

- **MGB OIL OUT** with MGB oil pressure below 5 psi and

 “GEARBOX OIL OUT”

- Increased power required at constant collective setting and airspeed
- Yaw kicks
- Unusual noise or vibrations
- Loss of two hydraulic pumps

- a. **LAND / DITCH IMMEDIATELY**; see [14.5 DITCHING / EMERGENCY LANDING](#)

----- END -----

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- 6. Land as soon as possible (a running landing is preferred)
- 7. Heater OFF
- 8. APU CTRL ON
- 9. APU GEN Confirm ON
- 10. Conditions permitting, descend below 500 feet AGL / ASL
- 11. Radio call Complete
- 12. Passengers Brief
- 13. Satellite communication Emergency
- 14. TCAS II [I] TA ONLY

----- END -----

Caution:

The main AC generators are cooled by MGB oil. Loss of oil or high oil temperatures may result in generator failure and loss of main electrical power. The APU should be started so the APU generator is available.

Considerations:

- If a leak occurs in the external cooler and **MGB OIL BYPASS** illuminates, the MGB oil temperature will increase (possibly rapidly depending on OAT) above 130°C (into the red)
- The MGB oil temperature sensors are wet bulb systems, which require some presence of oil to indicate properly. When MGB oil is lost, MGB oil temperature indications are not reliable. If the MGB oil pressure is below 5 psi, the MGB oil temperature indications should be ignored.
- A loss of main gearbox oil may result in the mechanical failure of a hydraulic pump drive
- There is a delay of 10 seconds between the oil pressure falling below 5 psi and the activation of the audio and visual warnings, to avoid false warnings triggered by pressure fluctuations with a single pump failure
- High oil temperature may cause the paint on the gearbox to smoke or emit fumes, which may be noticeable in the cockpit or cabin
- See [section 4 APU fuel consumption](#)

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5.8

MGB OIL OUT

Indications:

- MGB OIL OUT

 with

 "GEARBOX OIL OUT"

AND

MGB oil pressure below 5 psi

Actions:

1. LAND / DITCH IMMEDIATELY; see [14.5 DITCHING / EMERGENCY LANDING](#)

Warning:

A total loss of MGB oil pressure may result in MGB failure in less than 10 minutes.

----- END -----

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5.9

INPUT / ACC HOT

Indications:

- INPUT / ACC 1 HOT or INPUT / ACC 2 HOT

Actions:

1. Safe OEI flight Attain
2. Throttle (affected engine) IDLE, unless needed for safe flight
3. Land as soon as possible
4. Airspeed 80 to 100 KIAS
5. Pitch nose up (if practical) Maintain >2°
6. Heater OFF
7. APU CTRL ON
8. APU GEN Confirm ON
9. Consider descent to safe altitude
10. Radio call Complete
11. Passengers Brief
12. Satellite communication Emergency
13. TCAS II [I] TA ONLY
14. If throttle (affected engine) is at IDLE and engine is required for landing:
 - a. Throttle FLY on final

----- END -----

15. If IDLE engine is not required for landing:

- a. Good engine OEI 30S Confirm armed

----- END -----

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16. If ANY of the following symptoms of an imminent gearbox failure exist:

- **MGB OIL OUT** with

🔊 "GEARBOX OIL OUT"

- Increased power required at constant collective setting and airspeed
- Yaw kicks
- Unusual noise or vibrations
- Loss of two hydraulic pumps

- a. LAND / DITCH IMMEDIATELY;** see [14.5 DITCHING / EMERGENCY LANDING](#)

----- END -----

Considerations:

- See [section 4 APU fuel consumption](#)

5.10

MGB PUMP FAIL

Indications:

MGB PUMP 1 FAIL or MGB PUMP 2 FAIL

If: Oil pressure reduces below 45 psi with: MGB OIL PRESS followed by a further reduction below 35 psi with a change to: MGB OIL PRESS with audio "Gearbox pressure" and MGB OIL BYPASS • Possibly: INPUT / ACC HOT MGB OIL HOT with MGB temp above 130°C A MGB pump has failed See actions below.	If: No other indications A failure of a vacuum sensor pressure switch Actions: Land as soon as practical If other indications should appear, re-analyse ----- END -----	If: INPUT / ACC HOT illuminates within a few minutes – probable cause is: A leak in the vacuum sensing tube See actions below.
--	--	---

- Actions:**
- Safe OEI flight Attain
 - Throttle (affected engine) IDLE, unless needed for safe flight
 - If no indication of a leak is detected and ALL of the following indications exist:**
 - MGB PUMP 1 FAIL

 or

MGB PUMP 2 FAIL
 - MGB OIL BYPASS
 - MGB oil pressure is 7 psi or greater

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Caution:

There may be a drop of 2 psi in MGB oil pressure when the bypass switch is selected to MAN COOL (which is why the emergency procedure requires the pressure to be 7 psi or greater). If selecting MAN COOL causes the pressure to fall below 5 psi, reselect the switch to BYPASS.

- a. MGB OIL BYP switch MAN COOL

4. Land as soon as possible

5. Airspeed 80 to 100 KIAS

6. Pitch nose up (if practical) Maintain >2°

7. Heater OFF

8. APU CTRL ON

9. APU GEN Confirm ON

10. Consider descent to safe altitude

11. Radio call Complete

12. Passengers Brief

13. Satellite communication Emergency

14. TCAS II [I] TA ONLY

15. If throttle (affected engine) is at IDLE and engine is required for landing:

- a. Throttle FLY on final

----- END -----

16. If IDLE engine not required for landing:

- a. Good engine OEI 30S Confirm armed

----- END -----

17. If ANY of the following symptoms of an imminent gearbox failure exist:

- **MGB OIL OUT** with

🔊 “GEARBOX OIL OUT”

- Increased power required at constant collective setting and airspeed
- Yaw kicks
- Unusual noise or vibrations
- Loss of two hydraulic pumps

- a. **LAND / DITCH IMMEDIATELY**; see [14.5 DITCHING / EMERGENCY LANDING](#)

----- END -----

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Considerations:

- See [section 4 APU fuel consumption](#)

5.11

MGB OIL HOT

Indications:

- **MGB OIL HOT** with MGB oil temperature above 130°C

Note:

If **MGB PUMP 1 FAIL** or **MGB PUMP 2 FAIL** is illuminated, see [5.10 MGB PUMP FAIL](#).

If **INPUT / ACC 1 HOT** or **INPUT / ACC 2 HOT** is illuminated, see [5.9 INPUT / ACC HOT](#).

Warning:

If other indications, related to the MGB OIL pressure, comes on: Re-analyse.

Actions:

1. Land as soon as possible (a running landing is preferred)
2. APU CTRL ON
3. APU GEN Confirm ON
4. Check CBs IN
5. **MGB OIL BYP switch:**
 - a. MGB OIL BYP switch: BYP or MAN COOL
 - b. MGB OIL BYP switch: AUTO

Caution:

An increase in the MGB temperature will result in a slow MGB pressure drop to below the normal indication range. If pressure is still above 45 psi, reset the MGB OIL BYP switch only once.

6. Radio call Complete
7. Passengers Brief
8. Satellite communication Emergency

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9. If ANY of the following symptoms of an imminent gearbox failure exist:

- **MGB OIL OUT** with
 "GEARBOX OIL OUT"
- Increased power required at constant collective setting and airspeed
- Yaw kicks
- Unusual noise or vibrations
- Loss of two hydraulic pumps

a. **LAND / DITCH IMMEDIATELY**; see [14.5 DITCHING / EMERGENCY LANDING](#)

----- END -----

Caution:

The main AC generators are cooled by MGB oil. Loss of oil or high oil temperatures may result in generator failure and loss of main electrical power. The APU should be started so the APU generator is available.

Considerations:

- A hot air leak from the manifold area between the engines could cause 120°C hot air to be sucked in by the MGB oil cooler fan and prevent proper cooling of MGB oil
- See [section 4 APU fuel consumption](#)

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5.12

MGB CAUTION SYSTEM FAULT

Indications:

- M XMSN OIL WARN circuit breaker tripped (overhead panel aft row no.6 from right) and **MGB PUMP 1 FAIL** and **MGB PUMP 2 FAIL** and **MGB OIL HOT** and **MGB MAN COOL** will illuminate due to the loss of DC power to the associated switches and sensors. These are expected indications and do not indicate a MGB pump or cooling system failure.
- **Warning:**
If **MGB OIL PRES** warning is illuminated, MGB oil temperature exceeds 130° or MGB oil pressure drops below 35 psi, see [5.7 MGB OIL PRES – WARNING](#).

Actions:

1. Monitor MGB oil pressure and temperature on the EICAS (see considerations)
2. Land as soon as practical

----- END -----

Considerations:

- The MGB AUTO BYPASS system is not operational because the circuit breaker pop is perceived by the DCU as selecting MAN COOL. The **MGB MAN COOL** will be on regardless of the position of the MGB oil bypass switch or the position of the bypass valve. If MGB bypass is activated by switching to BYPASS, the **MGB OIL BYPASS** will illuminate.
- Should the red **MGB OIL PRES** illuminate, check the EICAS page to confirm MGB pressure. If the MGB pressure is below 35 psi, immediate manual activation of MGB bypass by selecting MGB OIL BYP switch to BYPASS will be necessary.

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5.13

MGB OIL PRESSURE CAUTION SYSTEM FAULT

Indications:

- M XMSN PRESS SWITCH circuit breaker tripped (panel right of throttles) and **MGB OIL PRES** and **MGB OIL BYPASS**
- MGB oil temperature increasing
- MGB oil pressure reducing by 5 to 10 psi (should indicate approximately 45-50 psi)
- **Warning:**
If **MGB OIL PRES** warning is illuminated, MGB oil temperature exceeds 130° or MGB oil pressure drops below 35 psi, see [5.7 MGB OIL PRES – WARNING](#).

Actions:

1. MGB OIL BYP switch MAN COOL
2. Monitor MGB oil pressure and temperature on the EICAS
3. Land as soon as practical

----- END -----

Considerations:

- An electrical short in the last jet switch or wiring will cause the M XMSN PRESS SWITCH circuit breaker to pop causing the **MGB OIL PRES** and **MGB OIL BYPASS** to illuminate and the MGB bypass valve to go into the bypass condition

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5.14

BYPASS TEST FAIL

Indications:

- **BYPASS TEST FAIL**

Actions:

1. Throttles FLY
2. MGB OIL BYP switch BYP or MAN COOL
3. MGB OIL BYP switch AUTO
4. If **BYPASS TEST FAIL** illuminates again:
 - a. Correct prior to flight
5. If **BYPASS TEST FAIL** does not illuminate:
 - a. Continue flight

----- END -----

----- END -----

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5.15

MGB MAN COOL

Indications:

- MGB MAN COOL

Actions:

1. If MGB MAN COOL was selected due to MGB pump failure:

- No additional action is required

----- END -----

2. If MGB MAN COOL was inadvertently selected:

- MGB BYP switch AUTO

----- END -----

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5.16

MGB OIL BYPASS

Indications:

- MGB OIL BYPASS

Note:

If MGB OIL PRES is illuminated, see [5.13 MGB OIL PRESSURE CAUTION SYSTEM FAULT](#).

Actions:

- If MGB pressure is less than 35 psi:
 - Confirm MGB OIL BYP switch AUTO
 - Continue with the appropriate MGB OIL PRESSURE / TEMPERATURE WARNINGS AND CAUTIONS emergency procedure

----- END -----

Note: 1

Abnormal flight conditions, such as severe turbulence or a low G pushover, may result in momentarily low MGB oil pressure and may inadvertently activate the automatic bypass system.

Note: 2

The MGB OIL BYPASS will latch if activated by the auto-bypass system. To unlatch the caution, select the MGB OIL BYP switch to BYP or MAN COOL then back to AUTO to rearm the system.

- If MGB pressure is greater than 35 psi and an abnormal flight condition can be identified:
 - MGB OIL BYP switch BYP or MAN COOL
 - MGB OIL BYP switch AUTO
- If the MGB OIL BYPASS activates again after arming the system without an abnormal flight condition:

- Remain in AUTO

----- END -----

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5.17 MGB SENSOR FAIL

Indications:

- MGB SENSOR FAIL

Actions:

1. Correct prior to flight

----- END -----

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T/RTR

6.1	LOSS OF TAIL ROTOR THRUST IN FORWARD FLIGHT
6.2	TAIL ROTOR CABLE FAILURE
6.3	TAIL ROTOR LOSS OF HYDRAULICS
6.4	TAIL ROTOR CONTROL MALFUNCTION IN FORWARD FLIGHT
6.5	LOSS OF TAIL ROTOR THRUST IN HOVER
6.6	LOSS OF TAIL ROTOR CONTROL IN HOVER

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6.1

LOSS OF TAIL ROTOR THRUST IN FORWARD FLIGHT

Indications:

- Uncontrollable yaw to the right, possibly preceded by excessive noise or vibration from the tail section or IGB CHIP or IGB HOT or TGB CHIP or TGB HOT

Actions:

1. Autorotation

Enter immediately (80 to 100 KIAS, 105% N_R)
2. APU CTRL

ON
3. Landing gear

Down
4. Radio call

Complete
5. If over water:

a. Floats

ARM

6. Throttles

STOP (before touchdown)

7. If time and altitude permit:

- a. Passenger

Brief
- b. Satellite communications

Emergency

----- END -----

6.2

TAIL ROTOR CABLE FAILURE

Indications:

- TR QUAD

Note:

If only one cable fails, no change in handling quality should occur.

Actions:

1. Power / airspeed Adjust to minimise yaw

2.

No response to pedals	Normal yaw control
See 6.4 TAIL ROTOR CONTROL MALFUNCTION IN FORWARD FLIGHT	Land as soon as practical

----- END -----

Considerations:

- With dual-cable failure, trimmed flight will occur at approximately 30 and 150 KIAS (these speeds vary with gross weight)

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6.3

TAIL ROTOR LOSS OF HYDRAULICS

Indications:

- **TR SVO 1 PRES** and **TR SVO 2 PRES**
- Possible **HYD 1 RSVR LOW** and **HYD 2 RSVR LOW**
- Yaw pedals may be stuck or restricted (unresponsive to pedal inputs)

Actions:

1. Power / airspeed Adjust to minimise yaw

2. See [6.4 TAIL ROTOR CONTROL MALFUNCTION IN FORWARD FLIGHT](#)

----- END -----

Considerations:

- Aircraft should fly in trim at about 60 KIAS and about 110 KIAS depending on weight. Aircraft will tend to yaw left between these speeds and yaw right either side of this speed range.

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6.4

TAIL ROTOR CONTROL MALFUNCTION IN FORWARD FLIGHT

Indications:

- Loss of response to yaw pedal inputs (may not be evident until a change of power setting)
- Yaw pedals may be stuck or restricted (unresponsive to pedal inputs)
- If **AFCS DGRD** illuminates and a BOOST, SAS, or TRIM hardover is suspected, see:
 - [9.11 BOOST HARDOVERS](#)
 - [9.12 SAS HARDOVERS](#)
 - [9.13 TRIM HARDOVERS](#)

Actions:

1. Power / airspeed	Adjust to minimise yaw
2. Yaw pedals	Centre (if possible). PM places feet on pedals to maintain central.

- Land as soon as possible
- Radio call Complete
- Passengers Brief
- Satellite communications Emergency
- TCAS II [\[I\]](#) TA ONLY
- APU ON
- Approach guidance for running landing:
 - Adjust power and airspeed as required to track towards a suitable airfield
 - On long final, reduce collective slowly with trim until nose yaws left
 - Reduce speed with cyclic pitch to achieve suitable rate of descent, keeping nose left of centre (aircraft may climb initially until speed is below V_Y)
 - Fly a normal approach profile, adjust airspeed to control ROD
 - Cushion touchdown with collective, if required. Nose will move right as collective is raised.
 - On ground, lower the collective slowly and use differential braking

----- END -----

Caution:

If during the approach a yaw rate to the right develops, go around using cyclic before further collective is applied.

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Note:

Certain failures will result in an aircraft response similar to a fixed pitch failure but with no response to left pedal input and limited response to right pedal input. During extended cruise flight, the pilot should limit the use of right pedal. For descent and landing, use available right pedal control as necessary.

Considerations:

- Any corrections to approach parameters should be small, using the trims, and made one control at a time
- With dual-cable failure, trimmed flight will occur at approximately 30 and 150 KIAS (these speeds vary with gross weight)
- With total loss of tail rotor hydraulics the aircraft should fly in trim at about 60 KIAS and about 110 KIAS depending on weight. Aircraft will tend to yaw left between these speeds and yaw right either side of this speed range.

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6.5 LOSS OF TAIL ROTOR THRUST IN HOVER

Indications:

- Uncommanded sharp right yaw, unresponsive to pedal inputs

Actions:

1. Collective

Lower to **LAND IMMEDIATELY**
2. Throttles

STOP
3. Brakes

Apply

----- END -----

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6.6

LOSS OF TAIL ROTOR CONTROL IN HOVER

Indications:

- Loss of response to yaw pedal inputs
- Yaw pedals may be stuck or restricted
- Uncontrollable yaw in either direction
- Possible **TR QUAD**

A. IF THE AIRCRAFT STARTS AN UNCONTROLLED YAW:

Actions:

1. Collective	Lower to LAND IMMEDIATELY
2. Throttles	STOP
3. Brakes	Apply

----- END -----

B. IF NO YAW IS INDICATED:

Actions:

- If the aircraft is over a suitable landing area:**
 - Collective Maintain setting
 - Attitude Maintain with cyclic
 - Throttles Reduce N_R to initiate a descent
 - After ground contact:**
 - Collective Set minimum pitch
 - Use differential braking to control heading

----- END -----

- If not over a suitable landing area:**
 - Attempt to flyaway using cyclic only
 - See [6.4 TAIL ROTOR CONTROL MALFUNCTION IN FORWARD FLIGHT](#)

----- END -----

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HYD

7.1	MAIN HYDRAULIC PUMP – SINGLE FAILURE
7.2	MAIN HYDRAULIC PUMP – DUAL FAILURE
7.3	NO.3 HYDRAULIC PUMP FAILURE
7.4	HYDRAULIC FANS FAILURE
7.5	HYDRAULIC SYSTEM OVER-TEMPERATURE
7.6	NO.3 HYDRAULIC SYSTEM LEAK
7.7	HYDRAULIC LEAK PAST TAIL ROTOR SHUTOFF
7.8	HYDRAULIC LEAK PRIOR TO TAIL ROTOR SHUTOFF
7.9	MAIN ROTOR SERVO PRESSURE
7.10	TAIL ROTOR SERVO PRESSURE
7.11	BOOST PRESSURE
7.12	SAS PRESSURE
7.13	LANDING GEAR WILL NOT RETRACT

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7.14	LANDING WITH GEAR RETRACTED
7.15	LANDING GEAR WILL NOT EXTEND
7.16	HARD LANDING

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7.1

MAIN HYDRAULIC PUMP – SINGLE FAILURE

Indications:

- **HYD 1 PUMP FAIL** or **HYD 2 PUMP FAIL**
- Affected hydraulic pressure indication below normal

Actions:

1. Hydraulic pressure gauges Monitor
2. Land as soon as practical

----- END -----

Considerations:

- With loss of a single main hydraulic pump, the no.3 hydraulic system will supply pressure to that system

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7.2

MAIN HYDRAULIC PUMP – DUAL FAILURE

Indications:

- **HYD 1 PUMP FAIL** and **HYD 2 PUMP FAIL**
- Both main hydraulic pressure indications below normal

Actions:

1. Control inputs Restrict to moderate rates
2. No.3 hydraulic pressure Monitor
3. Land as soon as possible
4. Radio call Complete
5. Passengers Brief
6. Satellite communication Emergency

----- END -----

Considerations:

- With loss of both primary hydraulic systems, no.3 hydraulic system will pressurise the primary servos allowing control of the aircraft

7.3

NO.3 HYDRAULIC PUMP FAILURE

Indications:

- **HYD 3 PUMP FAIL**
- No.3 hydraulic pressure gauge below red line

Actions:

1. Land as soon as practical
2. **On final approach:**
 - a. Landing gear handle DOWN
 - b. Landing gear EMER DN switch (<90 KIAS) Activate (hold down)
3. **After landing:**
 - a. EXT PWR Connect if available

----- END -----

Caution:

- a. Once the landing gear has been blown down, do not raise the gear handle until the gear has been inspected and serviced by maintenance personnel
- b. If there was a leak in the no.3 system, do not start the APU as it will cause the no.3 system to go to HI mode to try and recharge the accumulator. On the ground, this will also cause the ground pump to run dry.

Considerations:

- With the no.3 hydraulic system inoperative, the primary systems have no backup hydraulic pressure. The utility systems (landing gear and ramp) are no longer supplied and the landing gear must be blown down.
- Provided there has not been a leak in the no.3 hydraulic system (i.e. the cause of the fault is a pump failure), after landing, the ground pump can still be used to pressurise the utility system and lower the ramp

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7.4

HYDRAULIC FANS FAILURE

Indications:

- **HYDR FAN FAIL**

Actions:

1. Avoid extended hover and ground operations
2. Continue flight
3. Be prepared for **HYD TEMP ALERT** See [7.5 HYDRAULIC SYSTEM OVER-TEMPERATURE](#)

----- END -----

Considerations:

- Loss of both hydraulic cooling fans may cause one or both primary hydraulic systems to exceed normal temperatures, which will trigger the appropriate hydraulic temperature caution. Without cooling fans, the hydraulic system temperature will vary with ambient conditions. Any change in flight conditions may cause an increase or decrease in hydraulic system temperature.

7.5

HYDRAULIC SYSTEM OVER-TEMPERATURE

Indications:

HYD 1 TEMP ALERT

HYD 1 TEMP LIMIT

or

or

HYD 2 TEMP ALERT

or

HYD 2 TEMP LIMIT

or

or

HYD 3 TEMP ALERT

HYD 3 TEMP LIMIT

Actions:

1. If a TEMP ALERT illuminates:

- a. Land as soon as practical

----- END -----

2. If a TEMP LIMIT illuminates:

- a. Land as soon as possible
- b. Radio call Complete
- c. Passengers Brief
- d. Satellite communication Emergency

----- END -----

Considerations:

- Hydraulic temperatures may be checked on the HUMS page **HLTH** > **HUMS** > **MAINT** > **PARAMS** > **ENTER** > **ANLG**

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7.6

NO.3 HYDRAULIC SYSTEM LEAK

Indications:

- **HYD 3 RSVR LOW**

Actions:

1. Land as soon as practical
2. No.3 system pressure Monitor
3. If pressure drops below normal range or **HYD 3 PUMP FAIL** illuminates:
 - a. See [7.3 NO.3 HYDRAULIC PUMP FAILURE](#)

----- END -----

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7.7
HYDRAULIC LEAK PAST TAIL ROTOR SHUTOFF

Indications:

Illumination of either of the following sets of cautions:

TR SVO 1 PRES
HYD 1 RSVR LOW

or

TR SVO 2 PRES
HYD 2 RSVR LOW

Actions:

- Affected system pressure Monitor
- Land as soon as possible
- Radio call Complete
- Passengers Brief
- Satellite communication Emergency

----- END -----

Considerations:

- The LDI system has shut off the affected system backup isolation valve. Main servo pressure is unaffected and full control is still available.

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7.8

HYDRAULIC LEAK PRIOR TO TAIL ROTOR SHUTOFF

Indications:

Illumination of either of the following sets of cautions:

SAS 1 PRES	or	BOOST PRES
MR SVO 1 PRES		SAS 2 PRES
HYD 1 PUMP FAIL		MR SVO 2 PRES
TR SVO 1 PRES		HYD 2 PUMP FAIL
HYD 1 RSVR LOW		TR SVO 2 PRES
		HYD 2 RSVR LOW

- Affected hydraulic pressure gauge below normal
- Increased flight control forces (with no.2 system leak)

Actions:

- Maximum bank angle 30°
- V_{NE} 120 KIAS
- Avoid abrupt manoeuvring
- Land as soon as possible
- Radio call Complete
- Passengers Brief
- Satellite communication Emergency
- TCAS II [I] TA ONLY

----- END -----

Considerations:

- The LDI system has shut off the appropriate backup isolation valve and will not permit the no.3 system to pressurise the failed main system
- Order of caption illumination may vary depending on the leak rate

7.9

MAIN ROTOR SERVO PRESSURE

Indications:

- MR SVO 1 PRES or MR SVO 2 PRES

Actions:

1. SERVO switch Confirm to BOTH
2. If caution remains illuminated:

a. Maximum bank angle 30°
b. V_{NE} 120 KIAS
c. Avoid abrupt manoeuvring
d. Land as soon as possible
e. Radio call Complete
f. Passengers Brief
g. Satellite communication Emergency
h. TCAS II [I] TA ONLY

----- END -----

7.10

TAIL ROTOR SERVO PRESSURE

Indications:

- **TR SVO 1 PRES** or **TR SVO 2 PRES**

Actions:

1. SERVO switch Confirm to BOTH
2. **If caution remains illuminated:**
 - a. Land as soon as possible
 - b. Radio call Complete
 - c. Passengers Brief
 - d. Satellite communication Emergency

----- END -----

Note:

- a. If both **TR SVO 1 PRES** and **TR SVO 2 PRES** illuminated,
see [6.3 TAIL ROTOR LOSS OF HYDRAULICS](#)
- b. If any **HYD RSVR LOW** illuminates,
see [7.7 HYDRAULIC LEAK PAST TAIL ROTOR SHUTOFF](#)

7.11

BOOST PRESSURE

Indications:

- **BOOST PRES** with increased flight control forces

Actions:

1. HYD SERVO BST switch OFF then ON
2. **If caution light remains illuminated:**
 - a. HYD SERVO BST switch OFF
 - b. V_{NE} 120 KIAS
 - c. Land as soon as practical

----- END -----

Considerations:

- Coupled flight may be possible but it is unlikely that response will be adequate except in very calm conditions

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7.12

SAS PRESSURE

Indications:

- SAS 1 PRES and / or SAS 2 PRES

Actions:

- Affected SAS switch on AFCS panel OFF then ON
- If caution remains illuminated or returns:
 - V_{NE} 120 KIAS
 - Monitor the flight controls at all times
 - Land as soon as practical

----- END -----

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7.13

LANDING GEAR WILL NOT RETRACT

Indications:

- Red ● UNLKD light or one or more green lights illuminated at end of landing gear up cycle
- If AFCS ATT mode will not engage and radar, transponder, and TCAS remain in FSTBY, see [12.22 WEIGHT ON WHEELS INTERLOCK FAILURE](#)

Actions:

1. Airspeed 80 KIAS
2. L/G handle DOWN
3. **If landing gear extends correctly:**
 - a. Continue flight with landing gear down

----- END -----
4. **If landing gear will not extend:**
 - a. See [7.15 LANDING GEAR WILL NOT EXTEND](#)

----- END -----

Considerations:

- Once a safe down configuration is obtained after a landing gear malfunction, do not attempt to raise the gear unless required for safe flight
- If landing gear is extended, allow about 6% penalty in TAS at normal cruise power (with associated range penalty) and 50 feet/min on rate of climb OEI or 500 lb / 226 kg on the PC1 / PC2 weight (PC2E weight is unaffected)

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7.14

LANDING WITH GEAR RETRACTED

Caution:

With asymmetrical landing gear extension, the aircraft could roll over. Consideration should be given to evacuating all non-essential crew and passengers in a light-on-wheels hover before putting the full weight of the aircraft on the landing gear.

- Actions:**
- | | |
|------------------------|-------------------------------|
| 1. Radio call | Complete |
| 2. Suitable site | Select |
| 3. Fuel | Reduce to minimum |
| 4. Passengers | Brief |
| 5. Parking brake | ON |
| 6. Aircraft | Hover at approximately 2 feet |
| 7. Passengers | Evacuate (use ground crew) |
| 8. Floats | Arm / inflate (if required) |
| 9. Collective | Lower gently and fully |
| 10. Throttles | Retard slowly and fully |
| 11. Rotor brake | Apply cautiously |
| 12. Shutdown | Complete |

----- END -----

- Considerations:**
- If possible, arrange for suitable padding to be laid by ground staff. If not available, select soft ground or water landing site but consider implications of aircraft recovery.
 - Consider inflating floats in hover to assist ground stability. The forward floats will keep the nose of the aircraft off the ground but the rear float will not prevent the fuselage from grounding in the area of the ramp.

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7.15

LANDING GEAR WILL NOT EXTEND

Indications:

- Red ● UNLKD light ON or one or more green lights NOT illuminated at end of landing gear down cycle

Note:

The P/B DIMMER rheostat controls the bright and dim setting of the LDG GEAR control panel. In direct sunlight conditions, the LDG GEAR control panel lights will be DIM and appear off if the P/B DIMMER rheostat is not set in the OFF position. The bright setting is enabled only when the P/B DIMMER rheostat is in the fully OFF position.

Actions:

- P/B dimmer OFF
- P/B lamps Test (confirm bulb illuminates)
- If all green indicator lights do not illuminate during the test:**
 - Replace the malfunctioning indicator light with a functioning light and confirm proper gear indication
- If three greens illuminate on test and the red ● UNLKD light remains illuminated after the test:**
 - Airspeed 80 KIAS
 - L/G handle Cycle up then down
- If unsuccessful:**
 - Land as soon as practical
- On final approach:**
 - LDG GEAR handle Confirm down

Caution:

Ensure the LDG GEAR handle is in the DN position before activating the EMER DN switch.

- EMER DN switch (<90 KIAS) Activate (hold down)

Caution:

Once the landing gear has been blown down, do not raise the gear handle until the gear has been inspected and serviced by maintenance personnel.

- If red ● unlocked light remains illuminated and asymmetric gear extension is known or suspected:**
 - See [7.14 LANDING WITH GEAR RETRACTED](#)

----- END -----

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7.16

HARD LANDING

Indications:

- HARD LANDING

Actions:

- Shut down and hand over the aircraft to maintenance for required inspections

----- END -----

Considerations:

- The aircraft is not airworthy after a HARD LANDING
- Should the caution appear after a normal landing it could be caused by a faulty sensor switch
- Taxi with care if a faulty sensor switch is suspected

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8.1	AC GENERATOR – SINGLE FAILURE
8.2	AC GENERATOR – DUAL FAILURE
8.3	AC GENERATOR CONTROL UNIT FAILURE
8.4	BATTERY HOT
8.5	BATTERY CHARGE FAIL
8.6	CURRENT LIMITER FAILURES
8.7	DC CONVERTER – SINGLE FAILURE
8.8	DC CONVERTER – DUAL FAILURE
8.9	DC CONVERTER DUAL FAILURE AND BACKUP CONVERTER FAILURE
8.10	BACKUP CONVERTER OFF
8.11	BATTERY OFF

8.1AC GENERATOR – SINGLE FAILURE

Indications:

- AC GEN 1 FAIL or AC GEN 2 FAIL
- Possible momentarily outage of MFDs
- Possible loss of air data, AFCS DGRD, FD DGRD, AVC FAIL and INLT1/2 A-ICE FAIL while power is being reestablished from operative generator

Actions:

1. Heater OFF
2. APU CTRL ON
3. APU GEN Confirm ON
4. Failed generator switch RESET / OFF then ON
5. If unsuccessful:

a. Failed generator switch RESET / OFF

b. Land as soon as practical

----- END -----

Note:

If APU FAULT illuminates after single-generator failure, turn APU CTRL OFF unless required for safe flight.

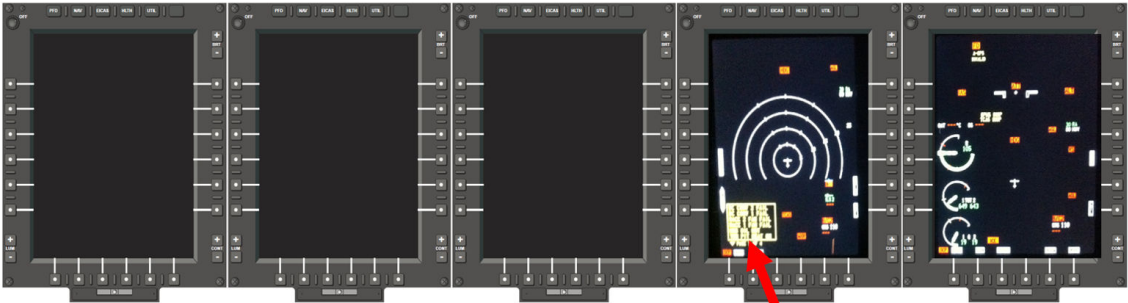
Considerations:

- See [section 4 APU fuel consumption](#)
- Systems that failed temporarily while power was being redirected may have to be reset

8.2

AC GENERATOR – DUAL FAILURE

Indications:

- 
 - LH MFDs go blank and RHP loses PFD info
 - AC GEN 1 FAIL** and **AC GEN 2 FAIL** will appear in the CAUTION, WARNING, and ADVISORY box at the bottom of the RH inboard MFD, on page 4; scroll down to display
 - AP and SSAS lost, both PSAS remain operative
 - RTU 1 is operational
 - Various MGB cautions (see considerations)

Actions:

- | | |
|----------------------------|--------------------------------|
| 1. Aircraft attitude | Control using STBY instruments |
| 2. APU CTRL | ON |

- | | |
|------------------|------------|
| 3. APU GEN | Confirm ON |
|------------------|------------|

- Note:**
If the APU GEN does not come on automatically, switch it to OFF / RESET then back to ON.
- | | |
|--|---------------------|
| 4. Failed generator switches (in turn) | RESET / OFF then ON |
|--|---------------------|

Continued on next page

5. If unable to reset, set generator switch to OFF
6. Did at least one main AC GEN reset or is AC available from the APU GEN?

Yes	No
7. MGB OIL BYP switch reset: a. MGB OIL BYP switch BYP or MAN COOL b. MGB OIL BYP switch AUTO 8. FMS ON 9. Land as soon as practical 10. If reset of both AC GEN was unsuccessful: a. Pilot windshield heat (only if required for safe flight) OVRD b. Icing conditions Vacate as soon as possible c. VMC Attain if possible	7. Attain VMC if possible 8. RH inboard MFD Select EICAS 9. Landing gear DOWN 10. EMER DN (below 90 KIAS) Activate 11. Land as soon as possible 12. Radio call Complete 13. Passengers Brief (Note: PA is inoperative) 14. See 12.18 RADIO TUNING UNIT FAILURE

----- END -----

Note:
Guard the collective firmly since the removal of AC will remove the trim force.

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Considerations:

● **If reset of AC is NOT successful:**

Battery is no longer charged. It will power the battery, holdup, and utility bus bars until it runs out of charge.

Consider turning battery off to conserve power. SAS will be lost as well as ICS / radios with BATT OFF.

The following MGB cautions will remain illuminated due to the loss of DC power from the essential bus to the M XMSN OIL WARN and M XMSN PRESS SWITCH circuit breakers:

MGB PUMP 1 FAIL and **MGB PUMP 2 FAIL** and **MGB OIL HOT** and **MGB OIL PRESS** and **MGB MAN COOL**.

● **If reset of AC is successful:**

With only the APU generator online, the RIPS main and tail blade heating elements, LH windshield anti-ice, LH engine inlet anti-ice, one air-conditioning condenser, and AVC are inoperative. The RIPS droop stop heaters remain powered. If pilot windshield heat is selected to OVRD, the **APU OVERLOAD** will illuminate.

During an in-flight AHRS alignment, the aircraft shall be flown straight and level for a period of at least 120 seconds. In turbulent air, it is possible that alignment will take longer than 120 seconds although the aircraft is flying straight and level. Attitude, heading, and VSI may become aligned at different times, depending on aircraft motion.

Set MFDs to include required systems (RADAR, TCAS, EGPWS).

- See [section 4 APU fuel consumption](#)

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8.3

AC GENERATOR CONTROL UNIT FAILURE

Indications:

- AC GCU 1 FAIL or AC GCU 2 FAIL

Actions:

- Heater OFF
- APU CTRL ON
- APU GEN Confirm ON
- Affected generator switch RESET / OFF then ON
- If unsuccessful:**
 - Failed generator switch RESET / OFF
 - Land as soon as practical

----- END -----

Considerations:

- Confirm all required systems are operating prior to continuing the flight
- AC GCU FAIL without the corresponding AC GEN FAIL could be caused by partial system failure, such as possible loss of under frequency protection. In this case AC GEN FAIL will illuminate on landing.
- See [section 4 APU fuel consumption](#)

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8.4

BATTERY HOT

Indications:

- **BATT HOT** and **BATT CHARGE FAIL**

Actions:

1. Land as soon as practical (see caution below)

----- END -----

Caution:

After flight with the battery automatically disconnected, there may not be sufficient battery power for subsequent APU starts. Shut down at a location where external power or battery charging capability is available.

Considerations:

- **BATT CHARGE FAIL** will accompany **BATT HOT**
- See [section 4 APU fuel consumption](#)

PRE

8.5**BATTERY CHARGE FAIL**

FIRE

Indications:

- **BATT CHARGE FAIL**

ENG

Actions:

1. Continue flight (however, see caution below)
2. Correct prior to next flight

APU

----- END -----

FUEL

Caution:

After flight with the battery automatically disconnected, there may not be sufficient battery power for subsequent APU starts. Shut down at a location where external power or battery charging capability is available.

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8.6

CURRENT LIMITER FAILURES

Indications:

- Various equipment failures without specific cautions

Note:

Tripping of the limiter between the AC bus bars will result in loss of the associated DC converter if an AC generator subsequently fails.

Actions:

1. Land as soon as practical

----- END -----

Considerations:

- Important equipment with single-power supply on DC essential bus: FMS 1, RH DCP, LG POS LTS, OEI PWR SEL, ADC 2, AFCS 2, FCC 2, VOR / ILS 2, trim coll, trim yaw, ramp control
- Important equipment with single-power supply on the battery bus: C/P ICS, VHF 1, fuel prime, HUMS, LG extend, pilot MWP, WOW 2

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8.7

DC CONVERTER – SINGLE FAILURE

Indications:

- **DC CONV 1 FAIL** or **DC CONV 2 FAIL**

Actions:

- Affected DC converter switch RESET / OFF then ON
- If unsuccessful:**
 - Land as soon as practical
- If scene lights required (SAR only):**
 - LOAD SHED OVRD OVRD
 - Nightsun will be lost (if installed)

----- END -----

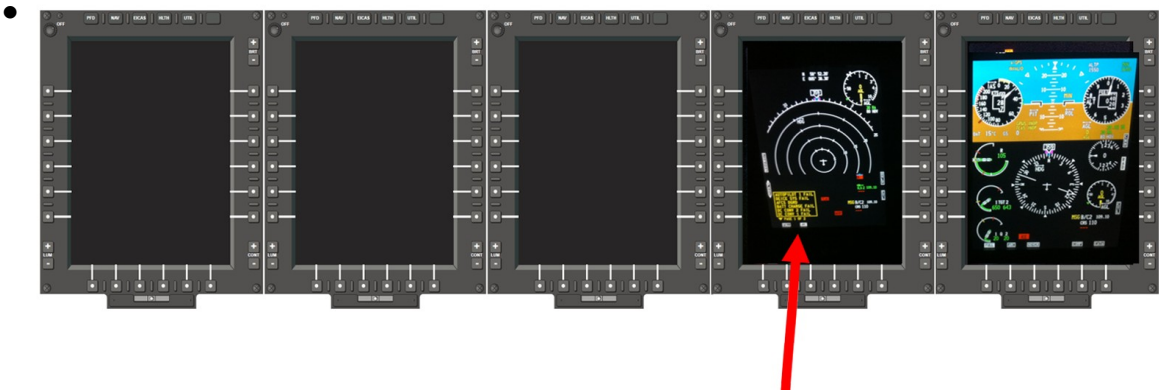
Considerations:

- When conducting the shutdown procedure, skipping the steps **(DC CONV 1 – OFF)** and **(DC CONV 1 – ON)** regardless of which DC converter may have failed, will retain all MFDs

8.8

DC CONVERTER – DUAL FAILURE

Indications:



- LH MFDs and LH MSP go blank
- **DC CONV 1 FAIL** and **DC CONV 2 FAIL** and **BATT CHARGE FAIL** and **AUTOPILOT 1 FAIL** will appear in the CAUTION, WARNING, and ADVISORY box at the bottom of the RH inboard MFD, on page 4; scroll down to display
-  **"DECOUPLE"**
- All DC primary bus systems lost
- RTU 1 is operational
- FMS 1 is operational

Actions:

1. BACKUP CONV switch Confirm to NORM
2. DC converter switches RESET / OFF then ON
3. **If neither converter recovered:**
 - a. Land as soon as practical
 - b. See [12.18 RADIO TUNING UNIT FAILURE](#)

----- END -----

Considerations:

- During an in-flight AHRS alignment, the aircraft shall be flown straight and level for a period of at least 120 seconds. In turbulent air it is possible that alignment will take longer than 120 seconds, although the aircraft is flying straight and level. Attitude, heading, and VSI may become aligned at different times, depending on aircraft motion.

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- **If reset of DC is NOT successful:**

FD is inoperative.

Flight controls will be reduced to RHS trim in pitch only. Loss of trim on LHS and in roll for RHS. SAS mode in roll despite ATT indicated on AFCS panel – treat as if in SAS mode.

Battery is no longer being charged. It will power the battery holdup and utility bus bars until it runs out of charge.

FD, RADALT, EGPWS, and TCAS are inoperative.

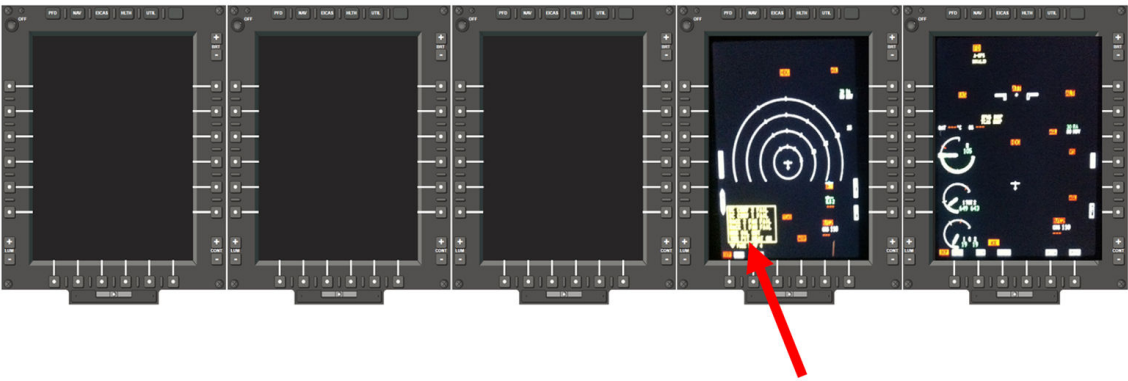
- **If reset of DC is successful:**

Set MFDs to include required systems (RADAR, TCAS, EGPWS).

8.9

DC CONVERTER DUAL FAILURE AND BACKUP
CONVERTER FAILURE

Indications:

- 
- LH MFDs go blank and RHP loses PFD info
- BACKUP CONV OFF** will appear in the CAUTION, WARNING, and ADVISORY box at the bottom of the RH inboard MFD, on page 4; scroll down to display. **DC CONV 1 FAIL** and **DC CONV 2 FAIL** cautions will illuminate after the backup converter has been turned off.
- Radios and **INTERCOM INOPERATIVE** until actions (step 2) is completed
- All DC primary bus, essential bus and battery bus systems lost
- SAS only

Caution:

V_{NE} 120 KIAS.

Note:

With only battery power, a series of cautions will illuminate to include the **MGB OIL HOT**, **AUTOPILOT 1 FAIL**, **AUTOPILOT 2 FAIL**, **AFCS DGRD**, **HYD FAN FAIL**, **RACK 1 FAN FAIL**, **RACK 2 FAN FAIL**, **INLT1 A-ICE FAIL**, **INLT2 A-ICE FAIL**, **DCU MISCOMPARE**, **AVC FAIL**, **CHIP SYS FAULT**, **MGB PUMP 1 FAIL**, **MGB PUMP 2 FAIL**, **MGB MAN COOL**, **MGB OIL PRESS**, **COCKPIT SEC FAN**, **DEICE SYS FAIL** and **BATT CHARGE FAIL** cautions. These are normal indications and do not signify a system malfunction.

Actions:

- | | |
|----------------------------|--------------------------------|
| 1. Aircraft attitude | Control using STBY instruments |
| 2. Backup converter | OFF |

Continued on next page

Note:

Battery bus is recovered.

3. Main converter switches (RESET) / OFF then ON

4. If at least one main converter is recovered:

- a. Land as soon as practical
- b. Backup converter NORM
- c. FMS ON

----- END -----

5. If unable to recover any converter:

- a. Attain VMC if possible
- b. RH inboard MFD Select EICAS
- c. Landing gear DOWN
- d. EMER DN (below 90 KIAS) Activate (hold down)
- e. Land as soon as possible
- f. Radio call Complete
- g. Passengers Brief (Note: PA is inoperative)

----- END -----

Note:

Battery is no longer charged. It will power the battery, holdup, and utility bus bars until it runs out of charge.

Considerations:

• **If reset of DC is NOT successful:**

The following MGB cautions will remain illuminated due to the loss of DC power from the essential bus to the M XMSN OIL WARN and M XMSN PRESS SWITCH circuit breakers:

MGB PUMP 1 FAIL and **MGB PUMP 2 FAIL** and **MGB OIL HOT** and **MGB OIL PRES**
and **MGB MAN COOL**

Consider turning battery off to conserve power. SAS will be lost as well as ICS / radios with BATT OFF.

Landing gear control panel lights will be inoperative.

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● **If reset of DC is successful:**

During an in-flight AHRS alignment, the aircraft shall be flown straight and level for a period of at least 120 seconds. In turbulent air it is possible that alignment will take longer than 120 seconds, although the aircraft is flying straight and level. Attitude, heading, and VSI may become aligned at different times, depending on aircraft motion.

Set MFDs to include required systems (RADAR, TCAS, EGPWS).

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8.10

BACKUP CONVERTER OFF

Indications:

- **BACKUP CONV OFF**

Actions:

1. BACKUP CONV switch NORM
2. **If caution light remains illuminated:**
 - a. Continue flight
 - b. Correct prior to next flight

----- END -----

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8.11

BATTERY OFF

Indications:

- **BATT OFF**
- Increased vibrations due to loss of AVC system

Actions:

1. Battery switch ON
----- END -----

Considerations:

- The following systems may fail:
 - Pilot and co-pilot windshield anti-ice
 - Number 1 engine anti-ice
 - Main and tail rotor blade de-ice
 - AVC system, consider adjusting speed for less vibrations
 - See [section SUPP.1 - 7](#) for electric overview

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9.1	AFCS DEGRADED
9.2	FD DEGRADED
9.3	FD ALTITUDE DEGRADED
9.4	AHRS VELOCITY DEGRADED
9.5	AHRS – SINGLE FAILURE
9.6	AHRS – DUAL FAILURE
9.7	PSAS FAULTS
9.8	AUTOPILOT – SINGLE FAILURE
9.9	AUTOPILOT – DUAL FAILURE
9.10	JAMMED OR RESTRICTED FLIGHT CONTROLS
9.11	BOOST HARDOVERS
9.12	SAS HARDOVERS
9.13	TRIM HARDOVERS

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9.14

FD FAILURE BELOW 50 KIAS

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9.1

AFCS DEGRADED

Indications:

- **AFCS DGRD**
- Possible uncommanded aircraft attitude change

Note:

- If one or more of the following data missing on any PFD: ASI, ALT, and / or VSI, see [12.1 ADC – SINGLE FAILURE](#)
- If heading and attitude information lost on one side of the cockpit, see [9.5 AHRS – SINGLE FAILURE](#)

Actions:

- Aircraft attitude Control
- If abrupt, uncommanded flight control movement in one or more axes with corresponding aircraft attitude change, see [9.11 BOOST HARDOVERS](#)

----- END -----
- If **PSAS 1 FAULT** or **PSAS 2 FAULT** illuminates, see [9.7 PSAS FAULTS](#)

----- END -----
- If slow, uncommanded flight control movement in one or more axes with corresponding aircraft attitude change, see [9.13 TRIM HARDOVERS](#)

----- END -----
- If aircraft attitude does not change:**
 - AFCS RESET (on control panel) Press
- If caution light remains illuminated or returns:**
 - Both AP PWR RESET (overhead panel) Press

Note:

Pressing the AP PWR RESET buttons will cause the aircraft to decouple and will momentarily put the AFCS into SAS mode.

- Continue flight Actively monitor controls

----- END -----

Considerations:

- Numerous possible causes. If system cannot be reset, faulty sensor can be determined from STATUS page (**HLTH** > **STATUS**).

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9.2

FD DEGRADED

Indications:

- FD DGRD
- Numerous possible causes, including improper sequencing of cockpit FD selector buttons

Actions:

- AFCS RESET (on control panel) Press
- If caution light remains illuminated or returns:**
 - Both AP PWR RESET (overhead panel) Press

Note:

Pressing the AP PWR RESET buttons will cause the aircraft to decouple and will momentarily put the AFCS into SAS mode.

- If caution light remains illuminated or returns:**
 - Continue flight using the other flight director

----- END -----

Note:

Do not use the faulty FD.

PRE

9.3

FD ALTITUDE DEGRADED

FIRE

Indications:

- **FD 1 ALT DGRD** or **FD 2 ALT DGRD**
- Certain systems faults will prevent the FMS or flight control computers from receiving correct altimeter setting data. This may degrade the accuracy of the FD ALT hold or FMS VNAV functions.

ENG

Actions:

1. Continue flight using the non-degraded flight director
2. **If both caution lights illuminate or ADC reversion is selected:**
 - a. Continue flight without using FD ALT mode or FMS VNAV functions

APU

FUEL

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----- END -----

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9.4

AHRS VELOCITY DEGRADED

FIRE

Indications:

- **AHRS 1 VEL DGRD** and / or **AHRS 2 VEL DGRD**
- Possible **A-GPS INVALID** on PFD above ASI, **TERRAIN INOP** on EGPWS display and "Be alert, terrain inop" audio alert

ENG

Actions:

APU

1. If only one AHRS velocity is degraded:

- Select the FD that corresponds to the healthy AHRS
- Continue flight

----- END -----

FUEL

2. If both AHRS velocities are degraded:

- Do not use FD below 50 KIAS
- Continue flight

----- END -----

HYD

Note:

- AHRS 1 / 2 VEL DGRD means the affected AHRS is not receiving valid information from the GPS
- Illumination of **A-GPS INVALID** is an indication that either the GPS signal has been completely lost or the connection between the FMS and the AHRS has been severed. A total loss of GPS signal will be accompanied by **TERRAIN INOP** and the audio alert. Without GPS aiding, the AHRS will continue to provide velocities for approximately 2 minutes.
- Degraded AHRS velocity will prevent using the affected flight director for the low airspeed SAR modes (MOT, VHLD, PHLD, CHOV, APP 2, ATPT, and DPT). This fault will not degrade coupled operations above 50 KIAS.

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9.5

AHRS – SINGLE FAILURE

Indications:

- Heading and attitude information lost on one side of the cockpit
- **AUTOPILOT FAIL** (affected side)
- Possible **AHRS VEL DGRD** (affected side) (SAR only)
- **AFCS DGRD**

Actions:

1. Reversion control panel Select operating AHRS
2. AP corresponding to failed AHRS (AFCS control panel) . Off
3. Select FD corresponding to valid AHRS
4. Continue flight in VMC
5. **If unable to maintain VMC:**
 - a. Land as soon as practical
 - b. V_{NE} 120 KIAS
 - c. See also MEL 2200-01 FCC and 3420-01 AHRS

----- END -----

Considerations:

- If AHRS 1 has failed:
 - a. MFD 1: Will lose the following information: Compass, ADI, VS
 - b. MFD 2: Will lose the following information: Compass
 - c. MFD 5: As MFD 1 (if in use as EICAS, nothing is lost)
 - d. WX radar will lose attitude control
- If AHRS 2 has failed:
 - a. MFD 4: Will lose the following information: Compass, ADI, VS
 - b. MFD 3: Will lose the following information: Compass

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9.6

AHRS – DUAL FAILURE

Indications:

- Heading and attitude information lost on both sides of the cockpit
- **AUTOPILOT 1 FAIL** and **AUTOPILOT 2 FAIL**
- Possible **AHRS VEL 1 DGRD** and **AHRS VEL 2 DGRD** (SAR only)
- **AFCS DGRD**

Actions:

1. Aircraft attitude Control using STBY instruments

2. Attain VMC as soon as possible
3. V_{NE} 120 KIAS
4. Land as soon as practical

----- END -----

Considerations:

- ADF may still be operative and available for navigation
- GPS position in FMS is still operative
- Aircraft position may be available in EFB
- WX radar will lose attitude control

9.7

PSAS FAULTS

Indications:

- PSAS 1 FAULT or PSAS 2 FAULT
- Possible uncommanded aircraft attitude change

Note:

If accompanied by same side SAS PRES, see 7.12 SAS PRESSURE.

A. In flight:

Actions:

1. If aircraft attitude changes:
- a. Aircraft attitude

Control
- b. See 9.12 SAS HARDOVERS
- END -----
2. If aircraft attitude does not change:
- a. AFCS RESET (on control panel)

Press
3. If caution light remains illuminated:
- a. Affected system secondary SAS

Select
- b. V_{NE}

120 KIAS
- c. Attain VMC as soon as practical
- d. Land as soon as practical
- END -----

Note:

Takeoff with a secondary SAS engaged is not authorised.

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B: On helideck or ground rotors running during a flight:

Actions:

1. AFCS RESET (on control panel) Press
2. **If caution light remains illuminated:**
 - a. Affected SAS Off
 - b. Continue flight in VMC

----- END -----

Considerations:

- See MEL 2210-02 'Primary stability augmentation system (PSAS)'

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9.8

AUTOPILOT – SINGLE FAILURE

Indications:

- **AUTOPILOT 1 FAIL** or **AUTOPILOT 2 FAIL** and **AFCS DGRD**
- Possible heading miscompare alert
- Possible uncommanded aircraft attitude change and / or flight control movement
- If abrupt, uncommanded flight control movement in one or more axes with corresponding aircraft attitude change, see [9.11 BOOST HARDOVERS](#)
- If **PSAS 1/2 FAULT** illuminates, see [9.12 SAS HARDOVERS](#)
- If slow, uncommanded flight control movement in one or more axes with corresponding aircraft attitude change, see [9.13 TRIM HARDOVERS](#)
- If coupled flight director will not engage and radar, transponder, and TCAS remain in FSTBY, see [12.22 WEIGHT ON WHEELS INTERLOCK FAILURE](#)
- If heading and attitude information lost on affected side of the cockpit, see [9.5 AHRS – SINGLE FAILURE](#)

Actions:

1. AFCS RESET (on control panel) Press
2. **If unsuccessful:**
 - a. Select FD to operative side
 - b. Failed AP PWR RESET Press
3. **If unsuccessful:**
 - a. Failed AP OFF
 - b. Continue flight

----- END -----

Considerations:

- Possible causes:
 - a. Heading miscompare, due to compass mismatch or AHRS failure. Check compass slaving, align if necessary.
 - b. Faulty trim or altitude / heading hold

9.9

AUTOPILOT – DUAL FAILURE

Indications:

- **AUTOPILOT 1 FAIL** and **AUTOPILOT 2 FAIL** and **AFCS DGRD**
- Possible heading miscompare alert
- Possible uncommanded aircraft attitude change and / or flight control movement
- If abrupt, uncommanded flight control movement in one or more axes with corresponding aircraft attitude change, see [9.11 BOOST HARDOVERS](#)
- If **PSAS 1/2 FAULT** illuminates, see [9.12 SAS HARDOVERS](#)
- If slow, uncommanded flight control movement in one or more axes with corresponding aircraft attitude change, see [9.13 TRIM HARDOVERS](#)
- If coupled flight director will not engage and radar, transponder, and TCAS remain in FSTBY, see [12.22 WEIGHT ON WHEELS INTERLOCK FAILURE](#)
- If heading and attitude information lost on both sides of the cockpit, see [9.5 AHRS – SINGLE FAILURE](#)

Actions:

1. Airspeed V_{NE} 120 KIAS
 2. AFCS RESET (on control panel) Press
 3. **If unsuccessful:**
 - a. AP 1 OFF
 - b. AP 2 - PWR RESET Press
 4. **If AP 2 RESET successful:**
 - a. FD 2 Select
 - b. AP 1 ON
 - c. See [9.8 AUTOPILOT – SINGLE FAILURE](#)
- END -----
5. **If AP 2 RESET unsuccessful:**
 - a. AP 1 ON
 - b. AP 2 OFF
 - c. AP 1 - PWR RESET Press
 6. **If AP 1 RESET successful:**
 - a. FD 1 Select
 - b. Continue flight

----- END -----

Continued on next page

7. If AP 1 RESET unsuccessful:

- a. AP 1 OFF
- b. Land as soon as practical

----- END -----

Considerations:

- Possible causes:
 - a. Heading miscompare, due to compass mismatch or AHRS failure. Check compass slaving, align if necessary.
 - b. Faulty trim or altitude / heading hold
- When the AP lights on the AFCS control panel are lit, the system is selected and powered. If electrical power is interrupted to the AP system, the AP lights will go blank.

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9.10

JAMMED OR RESTRICTED FLIGHT CONTROLS

Indications:

- Binding or restriction in cyclic or collective
- Possible **AFCS DGRD** or **FD DGRD** or **AUTOPILOT 1 FAIL** or **AUTOPILOT 2 FAIL**

Note: 1

If jammed pedals in forward flight is an indication, see also [6.4 TAIL ROTOR CONTROL MALFUNCTION IN FORWARD FLIGHT](#).

Note: 2

Trim actuators incorporate a slip clutch that provides an override capability in the event of a jammed trim actuator. The pilot applied forces required to overcome the slip clutch are approximately 15 lb in pitch, 12 lb in roll, and 30 lb in collective.

Actions:

1. Adjust flight controls to maintain aircraft control
2. Reduce airspeed to 120 KIAS or less
3. Check for foreign objects in the cockpit obstructing the flight controls
4. Trim OFF (affected control)
5. **If control continues to be restricted:**
 - a. Trim ON (affected control)
6. **If control continues to be restricted:**
 - a. SERVO BST OFF
7. **If control continues to be restricted:**
 - a. SERVO BST ON
8. **If control continues to be restricted:**
 - a. If possible, manually apply sufficient force to overcome restriction
 - b. Land as soon as possible
 - c. Radio call Complete
 - d. Passengers Brief
 - e. Satellite communication Emergency

----- END -----

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9.11

BOOST HARDOVERS

Indications:

- Possible **AFCS DGRD** or **FD DGRD**
- Abrupt, uncommanded flight control movement in one or more axes with corresponding aircraft attitude change

Actions:

1. Aircraft attitude Control
2. SERVO BST OFF
3. V_{NE} 120 KIAS
4. Attain VMC as soon as practical
5. Land as soon as practical

----- END -----

Considerations:

- Considerable force will be required to move the controls in one or more axes, even with the trim releases pressed
- With SERVO BST OFF coupled flight may be possible, but it is unlikely that response will be adequate except in very calm conditions

9.12

SAS HARDOVERS

Indications:

- Possible PSAS 1 FAULT or PSAS 2 FAULT or AFCS DGRD or AUTOPILOT 1 FAIL or AUTOPILOT 2 FAIL
- Abrupt, uncommanded aircraft attitude change in one or more axes without flight control movement

Actions:

1. Aircraft attitude Control
2. V_{NE} 120 KIAS
3. Faulty SAS OFF
4. Attain VMC as soon as practical
5. Land as soon as practical

----- END -----

Considerations:

- The aircraft will be noticeably less stable in one or more axes
- The TRIM, SAS and PRI / SEC lights on the AFCS control panel show switch position only. Illumination of these lights indicates that function is selected but does not ensure that function is operational. If electrical power is interrupted, these lights will remain illuminated.

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9.13

TRIM HARDOVERS

Indications:

- Possible **AUTOPILOT 1 FAIL** and / or **AUTOPILOT 2 FAIL** and / or **AFCS DGRD** and / or **FD DGRD**
- Slow, uncommanded aircraft attitude change in one or more axes together with corresponding flight control movement

Actions:

1. Aircraft attitude Control (use trim releases)
2. V_{NE} 120 KIAS
3. Affected trim system Disengage
4. Attain VMC as soon as practical
5. Land as soon as practical

----- END -----

Considerations:

- Following a cyclic trim failure, the pilot should still press and release the cyclic trim release button when established at a desired airspeed. This action will enhance airspeed stability.
- The SAS will initially counteract the hardover, but the attitude change will become rapid once the SAS actuators are saturated
- For flight with collective trim only disengaged, V_{NE} remains 165 KIAS
- Considerable force will be required to move the controls in one or more axes when initially taking control, but the force will disappear when the trim releases are pressed

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9.14

FD FAILURE BELOW 50 KIAS

Indications:

- AFCS DGRD
- Possible FD DGRD with "Decouple" audio alert
- Aircraft departs from desired flight path

Note:

Numerous possible causes. Switching FDs usually restores full functionality.

Actions:

1. If VMC:

- a. Maintain VMC
- b. Cyclic DECOUPLE button Press
- c. On AFCS control panel, other FD Select
- d. Reattempt coupled FD operations or continue uncoupled

----- END -----

2. If IMC:

- a. Cyclic DECOUPLE button Press
- b. Fly a manual go-around:
 - Set 5° nose-down, wings level
 - Increase power to establish positive rate of climb
 - At 50 KIAS transition to standard IMC climb parameters
- c. (At safe height / airspeed) Other FD Select
- d. Reattempt coupled FD operations or continue uncoupled above 50 KIAS

----- END -----

Considerations:

- If both FDs fail, continue in VMC (with adequate visual cues to provide a discernible horizon) or accelerate above 50 KIAS and continue the mission IFR. The coupled flight director, including SAR modes, will not be available after a dual-FD failure.

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A/ICE

10.1	PITOT HEAT FAILURE
10.2	TURN PITOT HEAT ON
10.3	TURN PITOT HEAT OFF
10.4	TURN ENGINE ANTI-ICE ON
10.5	ENGINE ANTI-ICE DEGRADED
10.6	ENGINE INLET ANTI-ICE FAILURE

10.1

PITOT HEAT FAILURE

Indications:

- PITOT 1 HEAT and / or PITOT 2 HEAT and / or PITOT 3 HEAT

Actions:

1. IAS or ALT modes (if engaged in pitch) Decouple
2. Affected pitot heat switch OFF then ON
3. If unsuccessful:
 - a. Affected pitot heat switch OFF
 - b. Select FD corresponding to operating PITOT HEAT (if any of 1 or 2)
 - c. Recouple as required
 - d. Continue flight avoiding icing conditions
4. If unable to vacate icing conditions:
 - a. Land as soon as possible
 - b. Radio call Complete
 - c. Passengers Brief
 - d. Satellite communication Emergency
5. If blocked Pitot is confirmed:
 - a. Reconfigure to valid ADC if necessary
 - b. Select FD corresponding to valid ADC
 - c. Recouple to IAS or ALT if required

----- END -----

Considerations:

- Possibility of pitot blockage. Cross-check other side and backup pitot-static indications. Airspeed indications may be unreliable with miscompare arrows on the ASIs. If airspeed indicates <50 KIAS the AP will no longer provide turn coordination during a turn but will revert to heading hold only. Manual pedal use for turn coordination may be necessary.
- VSI indications on the failed side will not be accurate unless the same side AHRS is also selected on the RCP
- Note that MEL 34 requires two operational Pitot static probes for flight in VMC only. For heating, see MEL 3030-01 'Pitot tube heaters'.
- A failure of PITOT 3 HEAT will not affect the coupling, continue as required

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10.2

TURN PITOT HEAT ON

Indications:

- TRN PIT HEAT ON
- OAT < +5°C with at least one pitot heat switch OFF

Actions:

1. All pitot heat switches ON

----- END -----

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10.3

TURN PITOT HEAT OFF

Indications:

- **TRN PIT HEAT OFF**
- Aircraft on ground, $N_R < 80\%$, at least one pitot heat switch ON

Actions:

1. All pitot heat switches OFF

----- END -----

- PRE
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10.4

TURN ENGINE ANTI-ICE ON

Indications:

- **TRN INLT A-I ON**
- $N_R > 80\%$, $OAT < +5^{\circ}C$, either engine inlet switch OFF

Actions:

1. Engine anti-ice switches Both ON

----- END -----

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10.5ENGINE ANTI-ICE DEGRADED

Indications:

- ENG 1 A-ICE DGRD or ENG 2 A-ICE DGRD

Actions:

- Icing conditions Vacate
- Continue flight outside icing conditions
- If unable to vacate icing conditions:**
 - Land as soon as possible
 - Radio call Complete
 - Passengers Brief
 - Satellite communication Emergency

----- END -----

Considerations:

- Anti-ice bleed air lost but electrical anti-ice still available

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10.6

ENGINE INLET ANTI-ICE FAILURE

Indications:

- INLT 1 A-ICE FAIL or INLT 2 A-ICE FAIL

Caution:

Failure of engine inlet anti-ice could cause loss of engine power.

Actions:

- Anti-ice switch (affected engine) OFF then ON (once only)
- If caution light remains illuminated:**
 - Icing conditions Vacate
 - Continue flight outside icing conditions
- If unable to vacate icing conditions:**
 - Land as soon as possible
 - Radio call Complete
 - Passengers Brief
 - Satellite communication Emergency

----- END -----

Considerations:

- Electrical anti-ice lost. Keep the switch ON to maintain anti-ice bleed air flow.

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RIPS

11.1	RIPS FAILURE
11.2	DROOP HEATER FAILURE
11.3	TURN RIPS ON
11.4	RIPS CONTROLLER FAILURE
11.5	PRIMARY MAIN ROTOR SLIPRING BEARING FAILURE
11.6	DUAL RIPS LWC SENSOR FAILURE
11.7	ICE DETECTED
11.8	RIPS MAIN ROTOR DEICE FAULT
11.9	DUAL RIPS OAT SENSOR FAILURE
11.10	INAPPROPRIATE RIPS MANUAL MODE
11.11	RIPS MAIN ROTOR DEICE FAILURE
11.12	RIPS TAIL ROTOR DEICE FAILURE
11.13	RIPS AUTO MODE FAILURE

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11.1

RIPS FAILURE

Indications:

- DEICE SYS FAIL
- ICE on both PFDs

Actions:

- Blade de-ice no.1 and no.2 reset switches Press and hold for approximately 3 seconds

Note:

If an inboard MFD is turned off or fails, the same side RIPS controller will not function until the MFD is rebooted. The other controller will provide RIPS control if functional.

Resetting the RIPS controllers will temporarily remove power from the system. The reset process can take 30 to 90 seconds, during which the RIPS will not be de-icing the rotor. If in icing conditions during a reset, the rotor will continue to accumulate ice during the reset process.

2. If normal RIPS function is restored:

- Continue flight in full or limited icing conditions at pilot's discretion

----- END -----

3. If normal RIPS function is not restored:

- Vacate icing conditions as soon as possible
- Avoid further flight in icing conditions

----- END -----

Considerations:

- Both RIPS controllers have failed causing a failure of the ice detection system

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11.2

DROOP HEATER FAILURE

Indications:

- **DROOP HTR FAIL**

Note:

Failure to heat the main rotor droops stops during icing conditions may result in a hung droop stop during rotor shutdown. A **DROOP HTR FAIL** can indicate a total or partial failure of one or more of the droop stop heaters. In the case of a partial failure, the RIPS will continue to heat the remaining droop stop.

Actions:

1. On shutdown See [12.16 HUNG DROOP STOP](#)

----- END -----

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11.3

TURN RIPS ON

Indications:

- TURN RIPS ON
- N_R >80%, ice detected, and RIPS OFF

Actions:

- RIPS AUTO or appropriate manual mode
- Engine inlet anti-ice ON

----- END -----

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11.4

RIPS CONTROLLER FAILURE

Indications:

- **DEICE CNTLR FAIL**
- **ICE** on PFD on failed side (LH for no.1 controller, RH for no.2 controller)

Note:

If an inboard MFD is turned off or fails, the same side RIPS controller will not function until the MFD is rebooted. The other controller will provide RIPS control if functional.

Actions:

1. Blade de-ice no.1 and no.2 reset switches Press and hold for approximately 3 seconds

Note:

Resetting the RIPS controllers will temporarily remove power from the system. The reset process can take from 30 to 90 seconds, during which time the RIPS will not be de-icing the rotor system. If flying in icing conditions during a controller reset, the rotor system will continue to accumulate ice during the reset process.

2. Continue flight whether or not controller is restored

----- END -----

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11.5 PRIMARY MAIN ROTOR SLIPRING BEARING FAILURE

Indications:

- **SLIPRING BRNG**

Actions:

1. Continue flight

----- END -----

Considerations:

- Secondary bearing will allow for continued operation for 10 hours

Indications:

- **RIPS DEGRD LWC**
- Ice rate meters on both PFDs no longer displayed

Note:

The RIPS will continue to sense OAT to schedule blade heating element ON time. However, without LWC input, the RIPS AUTO mode will schedule element OFF time based on a default LWC, which is equivalent to a mid-LIGHT icing level. If placed in a manual mode, the RIPS will schedule blade heating element OFF time based on the LWC associated with that particular manual mode and will continue to schedule blade heating element ON time based on actual OAT.

Caution:

A significant difference between the actual cloud LWC and the RIPS default LWC can lead to excessive ice accumulation (if the blades are not heated frequently enough) or run-back ice, which occurs when water from melted ice flows to an unprotected part of the blade and refreezes (if the blades are heated too frequently).

Actions:

- 1. If operating under limited icing clearance, go straight to step 4a**
- 2. If RIPS is operative:**
 - a. Monitor aircraft vibrations and power requirements
- 3. If aircraft vibrations or power requirements increase significantly:**
 - a. Select appropriate manual RIPS mode based on visual indications of ice accumulation or the system displayed ice rate prior to the failure

Note:

During normal system operation, aircraft vibrations (a vertical 2P under some conditions) and power requirements (up to approximately 15% torque change under some conditions) may cycle as the rotor blades accumulate and shed ice. Cyclical power requirements that peak higher than normal while in a manual mode may indicate that the next higher manual mode is more appropriate for the existing icing conditions. Steadily increasing, as opposed to cyclical vibration levels and power requirements, may be an indication of run-back ice.

- 4. If aircraft vibrations or power requirements continue to increase steadily:**
 - a. Vacate icing conditions as soon as possible
 - b. Avoid further flight in icing conditions

----- END -----

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11.7

ICE DETECTED

Indications:

- ICE DETECTED

Actions:

1. RIPS AUTO or appropriate manual mode
2. Engine inlet anti-ice ON

----- END -----

Considerations:

- When the RIPS is turned on, the ICE DETECTED will change to an ICE DETECTED advisory. The ICE DETECTED advisory will stay on as long as the RIPS senses ice accumulation and the RIPS is on.

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11.8

RIPS MAIN ROTOR DEICE FAULT

Indications:

- MR DEICE FAULT

Note:

One of the heating zones on the main rotor is malfunctioning, which may lead to abnormal or asymmetric ice accumulation or shedding. This may result in an unbalanced main rotor condition and increased vibration. Main rotor ice accumulation may also lead to increased power requirements.

Actions:

- Blade de-ice no.1 and no.2 reset switches Press and hold for approximately 3 seconds

Note:

Resetting the RIPS controllers will temporarily remove power from the system. The reset process can take from 30 to 90 seconds, during which time the RIPS will not be de-icing the rotor system. If flying in icing conditions during a controller reset, the rotor system will continue to accumulate ice during the reset process.

- If normal RIPS function is restored:

- Continue flight

----- END -----

- If normal RIPS function is not restored or if operating under limited icing clearance:

- Monitor aircraft vibrations and power requirements

- If aircraft vibrations or power requirements increase significantly:

- Vacate icing conditions as soon as possible
- Avoid further flight in icing conditions

----- END -----

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11.9

DUAL RIPS OAT SENSOR FAILURE

Indications:

- RIPS DEGRD OAT

Warning:

Without an OAT input, the RIPS OAT interlock will not function. In a manual mode, the RIPS will continue to heat the main and tail rotor blades regardless of OAT. Blade damage could occur at an OAT above +5°C. Therefore, use of a manual mode with a dual-RIPS OAT sensor failure is not recommended.

Note:

In all modes, the RIPS will schedule the blade heating element ON time by using a default OAT value of -12°C.

Caution:

A significant difference between the actual OAT and the RIPS default OAT can lead to excessive ice accumulation (if the blades are not heated long enough) or run-back ice (if the blades are heated for too long).

Actions:

- If operating under limited icing clearance, go straight to step 3a**
 - If RIPS is operative:**
 - Monitor aircraft vibrations and power requirements
 - Vacate icing conditions as soon as practical
 - If aircraft vibrations levels / power requirements steadily increase or cycle to abnormally high levels:**
 - Vacate icing conditions as soon as possible
- Note:**
- Selecting a RIPS manual mode will result in an immediate main and tail rotor heating cycle. Selecting a manual mode higher than the RIPS indicated LWC may help reduce power changes / requirements due to accumulation while exiting icing conditions.*
- After exiting icing conditions:**
 - RIPS AUTO or OFF
 - Avoid further flight in icing conditions

----- END -----

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11.10

INAPPROPRIATE RIPS MANUAL MODE

Indications:

- CHECK RIPS MODE

Note:

The selected RIPS manual mode does not match the cloud conditions as determined by the ice rate probes.

Actions:

- Select RIPS AUTO or appropriate manual mode based on the ice rate displayed by the ice rate meter on the PFD

----- END -----

Caution:

Operation of RIPS in a mode that does not match the actual conditions can lead to excessive ice accumulation or run-back ice.

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11.11

RIPS MAIN ROTOR DEICE FAILURE

Indications:

- MR DEICE FAIL

Note:

MR DEICE FAIL indicates two or more main rotor heating elements have failed. The system will initially react by assuming there is a communication loss between the primary RIPS controller and the blade heating elements and will automatically switch to the secondary controller. The switch may lead to a momentary flash of the MR DEICE FAIL as the secondary controller takes over and attempts to continue with normal system operation. If the fail condition still exists, the secondary controller will illuminate MR DEICE FAIL and shut down the remaining rotor de-ice zones. Go to HLTH > STATUS > RIPS to see the status of the blade heating elements and determine the extent of the main rotor failure.

Actions:

1. Monitor aircraft vibrations and power requirements

Note:

This fault may lead to abnormal ice accumulation. Consistently increasing (not cyclical) vibration levels and power requirements could indicate that the degraded RIPS is unable to de-ice the main rotor effectively.

2. If vibrations or power requirements increase significantly:

- a. Vacate icing conditions as soon as possible
- b. Avoid further flight in icing conditions

----- END -----

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11.12

RIPS TAIL ROTOR DEICE FAILURE

Indications:

- **TR DEICE FAIL**

Actions:

1. Blade de-ice no.1 and no.2 reset switches Press and hold for approximately 3 seconds

Note:

Resetting the RIPS controllers will temporarily remove power from the system. The reset process can take from 30 to 90 seconds, during which time the RIPS will not be de-icing the rotor system. If flying in icing conditions during a controller reset, the rotor system will continue to accumulate ice during the reset process.

2. If normal RIPS function is restored:

- a. Continue flight

----- END -----

3. If normal RIPS function is not restored or if operating under limited icing clearance:

- a. Monitor aircraft vibrations and power requirements

4. If aircraft vibrations or power requirements increase significantly:

- a. Vacate icing conditions as soon as possible
- b. Avoid further flight in icing conditions

----- END -----

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11.13

RIPS AUTO MODE FAILURE

Indications:

- AUTO DEICE FAIL**

Note:

Although the RIPS automatic mode has failed, the manual modes may still function properly. The pilot should use visual cues of ice accumulation or the ice rate displayed prior to the system failure to judge the proper manual mode for the existing conditions.

Caution:

Selection of the improper manual mode can lead to excessive ice accumulation (if the blades are not heated frequently enough) or run-back ice (if the blades are heated too frequently). Run-back ice occurs when water from melted ice flows to an unprotected part of the blade and refreezes.

Actions:

- Blade de-ice no.1 and no.2 reset switches Press and hold for approximately 3 seconds

Note:

Resetting the RIPS controllers will temporarily remove power from the system. The reset process can take from 30 to 90 seconds, during which time the RIPS will not be de-icing the rotor system. If flying in icing conditions during a controller reset, the rotor system will continue to accumulate ice during the reset process.

- If normal RIPS function is restored:**

- Continue flight

----- END -----

- If normal RIPS function is not restored:**

- Select appropriate manual RIPS mode

Note:

During normal system operation, aircraft vibrations (a vertical 2P under some conditions) and power requirements (up to approximately 15% torque change under some conditions) may cycle as the rotor blades accumulate and shed ice. Cyclical power requirements that peak higher than normal while in a manual mode may indicate that the next higher manual mode is more appropriate for the existing icing conditions. Steadily increasing as opposed to cyclical vibration levels and power requirements may be an indication of run-back ice.

- If cyclic power requirements peak higher than normal while in manual mode:**

- Select the next higher manual mode

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5. If normal RIPS function is not restored:

- a. Vacate icing conditions as soon as possible
- b. Avoid further flight in icing conditions

----- END -----

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12.1	ADC – SINGLE FAILURE
12.2	ADC – DUAL FAILURE
12.3	ADS-B OUT FAILURE
12.4	AVC SYSTEM MALFUNCTION
12.5	CHIP SYSTEM FAULT
12.6	CONFIGURATION MISCOMPARE
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12.9	DCU – MFD COMMUNICATION FAULT
12.10	DCU – SINGLE FAILURE OR MISCOMPARE
12.11	DCU – DUAL FAILURE
12.12	DOOR OPEN
12.13	CPI/AD ELT UNINTENTIONAL OPERATION

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12.14	FLOATS ARMED
12.15	FLOATS – INADVERTENT DEPLOYMENT
12.16	HUNG DROOP STOP
12.17	MAINTENANCE COMPUTER FAILURE
12.18	RADIO TUNING UNIT FAILURE
12.19	RAFTS ARMED
12.20	RAMP OPEN
12.21	ROTOR BRAKE
12.22	WEIGHT ON WHEELS INTERLOCK FAILURE

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12.1

ADC – SINGLE FAILURE

Indications:

- One or more of the following data missing on the affected side PFD: ASI, ALT, and / or VSI
- Possible **AFCS DGRD**

Actions:

- Reversion control panel Select operating ADC
- Select FD corresponding to valid ADC
- Reversion control panel Select same side AHRS
- Continue flight in VMC
- If unable to maintain VMC:**
 - Land as soon as practical

----- END -----

Considerations:

- VSI indications will be recovered when the same side AHRS is also selected on the RCP, if selected AHRS was to fail revert back to original AHRS
- WX radar input and wind data (ADC1). In the event of the failure of either the ADC1 or AHRS1, autotilt, stabilisation, and turbulence detection features will not be available but all other weather detection will be unaffected.

PRE

12.2**ADC – DUAL FAILURE**

FIRE

Indications:

- One or more of the following data missing on both sides PFD: ASI, ALT, and / or VSI

ENG

Actions:

1. Use standby altimeter and ASI for altitude and airspeed information. VSI indications will not be available.
2. Attain VMC as soon as possible
3. Land as soon as practical

APU

FUEL

----- END -----

PWRTN

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HYD

ELEC

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12.3

ADS-B OUT FAILURE

Indications:

- **ADS-B FAIL**
- For AMS 8.0:
 - a. On the RTU: **XPDR FAIL**
- For AMS 9.0:
 - a. Check PFD for XPDR or GPS fail indication
 - b. **ADS-B XPDR INOP** OR **ADS-B GPS INOP** on PFD

Actions:

1. Continue flight

----- END -----

Considerations:

- For AMS 8.0:
 - a. The specific fault is determined from the Transponder line on the MFD Health page by selecting HLTH > AVIONICS > SUBSTAT > TDR
 - b. If the transponder has failed, the line will show **MAINTENANCE** or **INOPERATIVE**
- a. If the wide area augmentation system (WAAS) position has failed, the line will show **OPERATIONAL**

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12.4

AVC SYSTEM MALFUNCTION

Indications:

- **AVC FAIL** or **AVC DEGRADED**
- Increased 4R vibration level

Actions:

- VIB CTRL switch OFF
- If vibration levels increase:**
 - VIB CTRL switch ON
 - If on ground perform an IBIT
 - Continue flight
 - Consider reducing airspeed to see if vibration levels reduce
- System reset procedure:**
 - VIB CTRL switch OFF
 - ANVC CMPTR PRI CB (no.1 panel, behind LH pilot, DC ESS) Pull
 - AVC CMPTR BU CB (BATT, overhead panel) (if installed) Pull
 - Wait 5 seconds
 - CBs Reset
 - VIB CTRL switch ON

----- END -----

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12.5

CHIP SYSTEM FAULT

Indications:

- CHIP SYS FAULT

Caution:

CHIP SYS FAULT indicates that the chip detector system is not able to detect metal particles in the gearboxes being monitored (MGB, IGB, TGB).

Actions:

- In flight:
 - Land as soon as practical

----- END -----
- On ground:
 - Do not takeoff until fault is corrected

----- END -----

- PRE
- FIRE
- ENG
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12.6

CONFIGURATION MISCOMPARE

Indications:

- CONFIG MISCOMP

Actions:

1. In flight:
- a. Continue flight
- END -----
2. On ground:
- a. Do not takeoff until fault is corrected
- END -----

Considerations:

- Generated by hardware or software mismatch between the MFDs (all MFDs must match), or between the DCUs (both sides must match), or the ADC OAT probes or AVC are not S92 versions

PRE

12.7

COOLING FAN FAILURE

FIRE

Indications:

- **COCKPIT SEC FAN** or **RACK 1 FAN FAIL** or **RACK 2 FAN FAIL**

ENG

Actions:

1. If necessary, decrease cockpit temperature using ECS
2. Continue flight

APU

----- END -----

FUEL

Note:

Both cooling fans in affected areas have failed.

PWRTN

T/RTR

HYD

ELEC

CTRL

A/ICE

RIPS

MISC

HOIST

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12.8

DCP DEGRADE OR FAILURE

Indications:

- **DCP** or **DCP**
- Possibly **FD DGRD**

Note:

A yellow **DCP** indicates that the information received from the on-side displays does not agree.

A red **DCP** indicates that the on-side displays are not receiving any data from the DCP.

Actions:

1. AFCS control panel Select FD corresponding to valid DCP

Caution:

Selecting the other FD will decouple all remaining FD modes.

2. AFCS RESET button Press
3. Mode select panel Couple desired FD modes
4. Attain VMC if possible
5. **If not possible:**
 - a. Land as soon as practical

----- END -----

Considerations:

- When a DCP fails, the primary NAV source on that side reverts to the active NAV frequency tuned on the corresponding RTU
- If a failure of the DCP occurs on the same side as the active flight director, the roll channel will decouple, if LNAV was engaged. In this case, it will generate a **FD DGRD**. Any attempt to change settings or make selections with the failed DCP will be unsuccessful.
- To adjust the altimeter subscale setting on the failed side, set the desired altimeter setting on the side with the serviceable DCP. Select the ADC reversion switch to the same side as the serviceable DCP. This will generate an **AFCS DGRD**. The altimeter setting will change to what is set on the serviceable side. Once the altimeter setting has been changed, the reversion switch should be placed back to NORM in order to retain ADC redundancy.

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CAS

12.9

DCU – MFD COMMUNICATION FAULT

Indications:

- CAS

or

CAS

Note:

A yellow

CAS

 indicates a communication problem between the DCU and MFD. In this case, the affected EICAS page is either not displaying all active warnings, cautions, or advisories (WCAs) or it is displaying false WCAs.

A red

CAS

 indicates a communication failure between the DCU and MFD. In this case, the affected EICAS page will not display most WCAs.

A. IN FLIGHT:

Actions:

- Select an EICAS page on the other side of the cockpit (one fed by the other DCU)
- If the alert is not shown on this other EICAS page:**
 - Continue flight, monitor the good EICAS

----- END -----

Note:

Selecting the non-affected DCU on the reversion control panel may or may not restore full WCA functionality to the affected EICAS.

- If the alert is shown on both sides of the cockpit:**
 - Land as soon as practical

----- END -----

B. ON THE GROUND:

Actions:

- Select an EICAS page on the other side of the cockpit (one fed by the other DCU)
- If the alert is not shown on this other EICAS page:**
 - Do not conduct flight in IMC until fault is corrected

----- END -----

Continued on next page

- PRE
- FIRE
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- ELEC
- CTRL
- A/ICE
- RIPS
- MISC
- HOIST
- LAND
- SUPP
- CAS

Note:

Selecting the non-affected DCU on the reversion control panel may or may not restore full WCA functionality to the affected EICAS.

3. If the alert is shown on both sides of the cockpit:

- a. Correct prior to flight

----- END -----

PRE

FIRE

ENG

APU

FUEL

PWRTN

T/RTR

HYD

ELEC

CTRL

A/ICE

RIPS

MISC

HOIST

LAND

SUPP

CAS

12.10

DCU – SINGLE FAILURE OR MISCOMPARE

Indications:

- **DCU MISCOMPARE** and
- **DCU 1** or **DCU 2** on EICAS
- For AMS 9.0: **DCU 1** or **DCU 2** on EICAS

Caution:

A DCU issue can cause inadvertent engagement of the main gearbox bypass valve. If the main gearbox pressure indications are >35 psi, and **MGB OIL BYPASS** is illuminated, cycle the main gearbox oil bypass switch to MAN COOL and back to AUTO to restore normal gearbox cooling.

Actions:

1. EICAS page Select on both sides of the cockpit (preferably MFD 3 and MFD 5, if installed, otherwise MFD 2 and MFD 3)
- Continued on next page

2. Determine if fault is a simple miscompare or DCU failure / degraded:

Cause	Indications	
DCU miscompare	- No EICAS data lost - Miscompare arrows next to discrepant parameters	
DCU degraded AMS 9.0	A degraded DCU 1 may cause some of the DCU 1 failure symptoms below.	A degraded DCU 2 may cause some of the DCU 2 failure symptoms below.
DCU failure AMS 8.0	DCU 1	DCU 2
1 (PFD)	Loss of N _R , N _P , TGT, and Q needles. Red dashes replace digital information.	Normal indications
2+5 (EICAS)	Loss of engine, XMSN, hydraulic, and fuel indications. Red dashes replace digital information.	FD DRGD FADEC 1 FAULT and FADEC 2 FAULT may also appear but with up to 30 seconds delay. DCU 2
3 (EICAS)	FD DRGD SAS 1 PRESS MGB OIL BYPASS HYD 3 PUMP FAIL SAS 2 PRESS MGB OIL PRESS FADEC 1 FAULT and FADEC 2 FAULT may also appear but with up to 30 seconds delay. Loss of hydraulic, MBG oil pressure and temperature indications. DCU 1	Loss of engine, XMSN, hydraulic, and fuel indications. Red dashes replace digital information. DCU 2 and CAS
4 (PFD)	Normal indications	Loss of N _R , N _P , TGT, and Q needles. Red dashes replace digital information.

A. IN FLIGHT:

Actions:

1. For DCU miscompare:

- a. Continue flight

----- END -----

Continued on next page

2. For DCU failure / degraded:

- a. Reversion control panel DCU switch Select operational DCU
- b. AFCS control panel Select FD corresponding to operational DCU

Note:

This is to ensure power limiting is available during coupled flight.

- c. Continue flight in VMC

3. If unable to maintain VMC:

- a. Land as soon as practical

----- END -----

B. ON THE GROUND:

Actions:

1. Do not conduct flight IMC until fault is corrected

----- END -----

Considerations:

- **DCU MISCOMPARE** is an indication that data from the two DCUs does not agree and is usually caused by differences between external sensors. Complete failure or internal fault of one DCU will also illuminate the **DCU MISCOMPARE**, and the appropriate DCU fail, or for AMS 9.0: **DCU** degraded indication. To differentiate between a simple miscompare and a DCU failure / degraded, compare EICAS on MFD 2 and MFD 3. A simple miscompare will display all data on both EICAS pages and may display dual-headed arrows on the discrepant parameters. For AMS 9.0 a **DCU** degraded may result in loss of some data.

12.11

DCU – DUAL FAILURE

Indications:

- **DCU 1** and **DCU 2** on EICAS
- No data on any EICAS display
- No N_R, N_P, TGT, or Q data on any PFD

or

For AMS 9.0: **DCU 1** and **DCU 2**

- Partial data presented on either side EICAS display
- Possibly a loss of any of the following: N_R, N_P, TGT, or Q data on either PFD

Caution:

A dual DCU failure may trigger activation of the MGB auto bypass. No CAS message, oil pressure or temperature indications will be presented to confirm this condition. Unless the dual DCU failure was preceded by a known and properly diagnosed oil leak in the MGB oil system, the MGB BYP switch should be placed in the MAN COOL position to allow proper MGB oil cooling.

Actions:

1.

MGB OIL BYP switch

MAN COOL
2.

Land as soon as possible
3.

Radio call

Complete
4.

Passengers

Brief
5.

Satellite communication

Emergency

----- END -----

Considerations:

- If both DCUs fail or are degraded, the performance of the aircraft will not be affected, however, the pilot will have limited ability to monitor aircraft health. Most cautions, warnings, and aural alerts will not be available. Attitude, heading, airspeed, and altitude displays will be available. Navigation and communication radios and displays will be available. Full AFCS functionality (ATT / SAS) will be available. FD collective modes will not be available.

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12.12 DOOR OPEN

Indications:

- **DOOR OPEN**

Actions:

1. Door Check visually
2. **If caution light remains illuminated:**
 - a. Land as soon as practical

----- END -----

PRE	12.13	CPI/AD ELT UNINTENTIONAL OPERATION
FIRE	Indications:	
ENG	- Possible beacon sound on 121.5 MHz - Possible E BEACON INOP advisory on EICAS - TX / TEST and / or BEACON GONE light on or flashing on ELT controller	
APU	Actions:	
FUEL	1. TEST / RESET button Press	
PWRTN	2. TX / TEST light Confirm out	
T/RTR	3. If necessary:	
HYD	a. ATC Inform	
ELEC	b. Continue flight	
CTRL	4. If transmission cannot be stopped, inform engineering on next landing	
A/ICE	----- END -----	
RIPS		
MISC		
HOIST		
LAND		
SUPP		
CAS		

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12.14

FLOATS ARMED

Indications:

- FLOATS ARMED

Actions:

- Floats SAFE
- If caution still illuminated:
 - Airspeed Reduce below 80 KIAS

----- END -----

Note:

FLOATS ARMED caution is generated with floats armed in the air at speeds above 80 KIAS or on the ground with N_R below 80%. At airspeeds at or below 80 KIAS, a FLOATS ARMED advisory will be displayed.

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12.15

FLOATS – INADVERTENT DEPLOYMENT

Indications:

- Possibly a loud bang followed by float bags appearing outside the pilot and co-pilot windows

Actions:

1. Maximum rate of climb 1000 feet/min
2. Maximum bank angle 30°
3. V_{NE} 80 KIAS calibrated airspeed

Note:

Deployed floats adversely affect the accuracy of the airspeed system. The IAS shown below are equivalent to 80 KIAS calibrated airspeed.

Maximum IAS with floats deployed	
Flight condition	Maximum KIAS
Climb	50
Level flight	55
Descent / autorotation	60

4. Vacate / avoid icing conditions
5. Land as soon as practical

----- END -----

Caution:

- a. Avoid flight in icing conditions with floats deployed
- b. Due to decreased airspeed and increased drag, aircraft range will be severely limited with floats deployed
- c. Restrict nose-up flare attitude on touchdown to 8° or less to avoid hitting the aft float bag

PRE

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12.16

HUNG DROOP STOP

INDICATIONS

- One or more droop stops not in on shutdown; affected blade(s) flying much lower than normal at IDLE N_R

Warning:

Ensure all personnel are well clear of the rotor disc when stopping the rotor with a droop stop not engaged.

Actions:

- One engine Shut down to reduce N_R
- If droop stop(s) remains out:**
 - N_R Increase to above 75%
 - Slightly displace cyclic to try and dislodge the droop stop
 - N_R Decrease to IDLE
- If unsuccessful after several attempts:**
 - Remaining engine Shut down
 - Cyclic Neutral
 - If conditions permit, do not apply the rotor brake

----- END -----

Considerations:

- Utilise qualified ground personnel to confirm droop stops engaged
- If qualified ground personnel are not available and shutdown is necessary, consider utilising LHP to visually check droop stops engaged before shutdown

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12.17

MAINTENANCE COMPUTER FAILURE

Indications:

- MAINT COMP FAIL

Actions:

- Continue flight
- Consult engineering on return to maintenance base
- Manually record any engine or MGB exceedances, as the HUMS will not be available
- If starting APU:**
 - Fuel prime PUMP 1 or PUMP 2

Note:

MAINT COMP FAIL may take up to 2 minutes to clear after BATT ON.

- System reset procedure. NB pull both circuit breakers simultaneously:**
 - MDC primary CB (no.2 DC pri, top row) Pull
 - MDC secondary CB (BATT HOLD UP, overhead panel) Pull
 - Wait 5 seconds
 - CBs Press in

----- END -----

Considerations:

- PARKING BRAKE advisory will not be displayed if the parking brake is applied
- Advisories that are displayed when the maintenance computer fails might not extinguish as normal

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12.18

RADIO TUNING UNIT FAILURE

Indications:

- One RTU partially or fully inoperative

Actions:

1. Failed RTU OFF
2. Operative RTU Select 1 / 2
3. Cross-side radios / nav aids Tune from operative RTU
4. Continue flight

----- END -----

Note:

Cross-side radio / nav aid labels are shown in yellow on the on-side RTU.

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- SUPP
- CAS

12.19

RAFTS ARMED

Indications:

- RAFTS ARMED

Actions:

1. Raft SAFE

2. If caution light extinguishes:

- Continue flight

----- END -----

3. If caution light remains illuminated:

- Land as soon as possible
- Radio call Complete
- Passengers Brief
- Satellite communication Emergency

----- END -----

PRE

FIRE

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ELEC

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HOIST

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CAS

12.20

RAMP OPEN

Indications:

- RAMP OPEN

Actions:

1. Ramp switch [I] UP
2. Cameras [I] Check on MFD
3. If caution light does not extinguish:
 - a. Land as soon as practical

----- END -----

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12.21

ROTOR BRAKE

Indications:

- ROTOR BRAKE

A. IN FLIGHT:

Actions:

1. Rotor brake lever Ensure fully up in detent
2. **If lever positioned correctly and caution light remains illuminated:**
 - a. Investigate for fire, vibrations, or unusual noise
 - b. Land as soon as possible
 - c. Radio call Complete
 - d. Passengers Brief
 - e. Satellite communication Emergency
3. **If signs of fire appear:**
 - a. **LAND / DITCH IMMEDIATELY;** see [14.5 DITCHING / EMERGENCY LANDING](#)

----- END -----

B. ON THE GROUND:

Actions:

1. Rotor brake lever Ensure fully up in detent
2. **If lever positioned correctly and caution light remains illuminated:**
 - a. Shut down and investigate

----- END -----

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12.22

WEIGHT ON WHEELS INTERLOCK FAILURE

Indications:

- Landing gear will not retract (red travel light on with lever up)
- AFCS ATT mode will not be operative even though the green ATT light may be on
- Radar, transponder, and TCAS may remain in FSTBY
- RIPS will be inoperative although not indicated as such (although a heating cycle will be possible using the OVRD command)
- Unable to couple to any flight director
- Possible **AUTOPILOT 1 FAIL** or **AUTOPILOT 2 FAIL**

Actions:

- Note:**
- Before WOW CBs are pulled:
- It will be possible to test the CVR / FDR
 - The transponder code display will be green, however, ATC will not be able to receive the transponder transmission
 - Radar and digital clock wired to the RH WOW switch only. If they are operative, only the LH WOW switch interlock has failed.
 - Electrical system protections such as AC generator under frequency or DC system under-voltage will not be available
- WOW no.1 CB Pull (overhead panel, BATT HOLD UP bus, bottom row, far left with yellow collar)
 - WOW no.2 CB Pull (overhead panel, BATT bus, second row, fifth from right B16, middle with yellow collar)
 - Land as soon as practical
- END -----
- On the ground:**
 - Both CBs Reset

----- END -----

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HOIST

13.1	HOIST OVERTEMP
13.2	HOIST CONTROL FAILURE
13.3	HOIST CONTROL RUNAWAY
13.4	HOIST EMERGENCY CHANGEOVER
13.5	HOIST EMERGENCY RECOVERY OF THIRD PARTY ON HOIST

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13.1

HOIST OVERTEMP

Indications:

- **OVERTEMP** lights illuminate on the cabin rescue hoist panel and the crewman's hoist pendant

Actions:

1. Hoist operator Inform PF
 2. Aircraft Manoeuvre to a safe position
 3. If someone is on the wire Make safe as quickly as possible

Note:

'Make safe as quick as possible' means to recover to the aircraft or place back to the location they came from whichever is deemed the safest course of action. See OMF for more information.

4. Use alternate hoist if mission critical

----- END -----

Considerations:

- The affected hoist will run at 60% of normal speed when the **OVERTEMP** light is on

PRE

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13.2

HOIST CONTROL FAILURE

Indications:

- Hoist does not respond to pendant controls
- Hoist cockpit and cabin switch position – normal
- Flashing **OVERTEMP** light or alternating flashing **OVERTEMP** and caution lights on hoist pendant and / or hoist operator control panel

CASE A: Personnel on the hook

Actions:

1. Hoist operator

Inform PF
2. Aircraft

Manoeuvre to a safe position

3. For dual hoist installation only:

RCM on wire

- a. Consider lowering the RCM and casualty to a safe area using the aircraft's height control (i.e. closest safe point of land to place the RCM on the ground or vessel if conditions permit). If not able to complete, go to b.
- b. Continue to

See [13.4 HOIST EMERGENCY CHANGEOVER](#)

If third party on wire – see [13.5 HOIST EMERGENCY RECOVERY OF THIRD PARTY ON HOIST](#)

4. If single hoist operation or secondary hoist does not respond to pendant control:

- a. Consider lowering the personnel to a safe area using the aircraft's height control (i.e. closest safe place land / vessel)

5. If unable:

- a. EICAS

Confirm **HOIST ARMED** advisory on
- b. Backup pendant
- c. Hoist control panel
- d. PF cyclic switch
- e. PM cyclic switch

----- END -----

Continued on next page

- PRE
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CASE B: No personnel on the hook

Actions:

1. Check alternative options / controls

- a. EICAS
- b. Backup pendant
- c. Hoist control panel
- d. PF cyclic switch
- e. PM cyclic switch
- f. Consider lowering load to ground using aircraft
- g. Recover cable by hand into cabin

Confirm **HOIST ARMED**
advisory on

----- END -----

13.3

HOIST CONTROL RUNAWAY

Indications:

- Hoist does not respond to pendant control

Actions:

1. Hoist operator

Inform PF
2. EMERGENCY STOP button

Press

- Hoist operator confirms to PF that the hoist has stopped
3. Aircraft

Manoeuvre to a safe position

4. If no person on the wire: Attempt to reset the EMERGENCY STOP button once only

5. For dual hoist installation only:

If problem not resolved

- a. Continue to See [13.4 HOIST EMERGENCY CHANGEOVER](#)

If third party on wire – see [13.5 HOIST EMERGENCY RECOVERY OF THIRD PARTY ON HOIST](#)

----- END -----

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13.4
HOIST EMERGENCY CHANGEOVER

Consider lowering the RCM and casualty to a safe area using the aircraft's height control (i.e. closest safe point of land to place the RCM on the ground or vessel if conditions permit).

Actions:

- Press active hoist button**
 - If proceeding from [13.3 HOIST CONTROL RUNAWAY](#) the EMERGENCY STOP button will be pressed in. The EMERGENCY STOP button will need to be reset to restore power to the alternate hoist, after the active hoist button has been pressed.
- Attach a 'quickdraw' between the hook of the serviceable hoist and the cable of U/S hoist
- Lower serviceable hoist hook to the RCM, using U/S cable as a guide
- RCM attaches serviceable hoist hook through lower D ring on QRF or karabiner (if in use) and gives a double thumbs up when attached
- Hoist operator winches in approximately one foot to take the RCM's weight on serviceable hoist
- RCM will then disconnects from U/S hoist hook, holds it clear and gives a thumbs up
- Recovery of the RCM**
 - Recover the RCM to the aircraft ensuring that the U/S cable / hook do not come into contact with the ground / water
- If a SIMULATED hoist failure:**
 - Switch back to original hoist and recover cable
- Once the RCM is safely inboard the aircraft, secured, and the serviceable hoist is fully winched in:**
 - Recover the U/S cable by hand and then cut it approximately 3 inches from the hoist assembly using the manual cable cutters only
 - If unable to recover the U/S by hand, conduct the aircraft to a safe area and cut the U/S cable using the **manual cable cutters only**

----- END -----

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13.5

HOIST EMERGENCY RECOVERY OF THIRD PARTY ON HOIST

Consider lowering the third party to a safe area using the aircraft's height control, i.e. closest safe point of land to place third party on the ground or vessel if conditions permit, taking into consideration the time that the third party would have to stay on the strop / ARV.

Actions:

1. Press Active Hoist button
- a.

If proceeding from [13.3 HOIST CONTROL RUNAWAY](#), the EMERGENCY STOP button will be pressed in. The EMERGENCY STOP button will need to be reset to restore power to the alternate hoist, after the active hoist button has been pressed.
2. Hoist operator connects the KHG to the serviceable hook and attaches it to the u/s hoist cable. Hoist operator ensures the swing latch is secured by a cable tie.
3. Hoist operator lowers KHG to the bumper plate of the u/s hoist cable
4. Hoist in and recover the third party to the aircraft ensuring the slack cable is managed
5. The third party must be placed aft of the dam before disconnecting the KHG
6. Once the third party is secured, stow the hoist in use
7. Use the manual cable cutters only to cut the U/S hoist cable 3 inches from the hoist assembly
8. Recover the crewman on the remaining hoist or continue with mission

Note:

During hoisting operations with KHG, the helicopter should adjust its altitude downward while reeling in the hoist cable. This is done to maintain the least possible distance from the water / ground for the person suspended on the hoist, in case any unforeseen event occurs with the winch system.

----- END -----

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LAND

14.1	LIGHTNING STRIKE
14.2	BOMB THREAT
14.3	PILOT INCAPACITATION
14.4	PASSENGER BRIEFINGS
14.5	DITCHING / EMERGENCY LANDING
14.6	EMERGENCY EVACUATION ON WATER
14.7	EMERGENCY EVACUATION ON LAND

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14.1

LIGHTNING STRIKE

Indications:

- Flash and / or bang
- Possible static noise in the intercom and radios
- Possible failure of avionics equipment
- Possible erratic compass and NDB indications
- Possible increase in vibration levels
- Possible 'ozone' smell

Actions:

1. Land as soon as possible
2. Radio call Complete
3. Passengers Brief
4. Satellite communication Emergency
5. **If vibration increases significantly:**
 - a. **LAND / DITCH IMMEDIATELY;** see [14.5 DITCHING / EMERGENCY LANDING](#)

----- END -----

Considerations:

- If the aircraft has suffered a lightning strike, it is likely there will be damage to the main and / or tail rotor blades, rotor heads and associated components. The majority of this damage may not be visible without strip examination. Once the aircraft has landed after a lightning strike, it shall not be flown again until it has been fully inspected by appropriate technical personnel.

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- SUPP
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14.2

BOMB THREAT

- Actions:**
1. Radio call Complete
 2. Divert to nearest suitable landing site
 3. Satellite communication Emergency
 4. **After landing:**
 - a. Passengers Evacuate to safe location
 - b. Aircraft Shut down
- END -----

- Considerations:**
- For further guidance, see section 'Bomb threat to a Company aircraft' in OMA chapter 10

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14.3

PILOT INCAPACITATION

Indications:

- Sudden or progressive incapacitation should be suspected if:
 - PF fails to maintain assigned heights or headings and / or makes exaggerated flight path corrections or alterations
 - Either pilot begins to deviate from normal procedures
 - Either pilot fails to respond to a second verbal communication
 - Either pilot's speech becomes slurred

Actions:

- Take control of the aircraft and establish or maintain a safe flight path
- Clearly announce **"I have control"**
- If available, request rear / cabin crew assistance to secure the incapacitated pilot in their seat with arms positioned inside the shoulder harness. Do not attempt to remove the pilot from their seat.
- Make use of FD modes when possible
- If available, request rear / cabin crew to render first aid to incapacitated pilot. This action shall not jeopardise safe flight.
- Select the nearest suitable landing site considering:**
 - Present position
 - En route and destination weather
 - Fuel endurance and fuel at destination
 - ATC and availability of radar assistance
 - Condition of incapacitated pilot
 - Availability of medical assistance on board or at destination
- Radio call – as required
- Transponder – as required
- [I] ISAT – EMERG
- Fly to destination:**
 - Get crew (if available) to assist with navigation, approach plates, and checklists
 - Request medical assistance on arrival

----- END -----

DITCHING BRIEFING:**Note:**

May be included within the automatic passenger briefing system, if equipped.

- a. **"This is a ditching warning"**
- b. **"Prepare for ditching. I repeat, prepare for ditching."**
- c. [I] **"Do not inflate life-jackets until outside aircraft"**
- d. **"Check your seatbelt is fastened securely"**
- e. **"Remove glasses, dentures, and sharp objects"**
- f. **"Take up brace position on pilot's signal"**
- g. **"After landing, follow crew's instructions"**

----- END -----

EMERGENCY LANDING BRIEFING:**Note:**

May be included within the automatic passenger briefing system, if equipped.

- a. **"This is an emergency landing warning"**
- b. **"Prepare for emergency landing"**
- c. **"I repeat, prepare for emergency landing"**
- d. **"Check your seatbelt is fastened securely"**
- e. **"Remove glasses, dentures, and sharp objects"**
- f. **"Take up brace position on pilot's signal"**
- g. **"After landing, follow crew's instructions"**

----- END -----

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14.5

DITCHING / EMERGENCY LANDING

Actions:

- Collective Maintain min 15% torque
 - Descend Recommended 80 KIAS
 - HDG bug Set to wind direction
 - Autopilot If possible, select 3 cue using ALTP, HDG, and IAS
 - VS Adjust depending on the type of emergency
 - Heater OFF
 - APU CTRL ON
 - APU GEN Confirm ON
 - Radio call Complete
 - Satellite communication Emergency
 - Landing gear DOWN
 - Floats (over water) Arm below 80 KIAS
 - Passengers Brief
 - 500 feet RADALT Reduce rate of descent and select RALT and beep down, adjust speed to 55 KIAS
- 15. If one engine throttle was retarded to IDLE:**
- Throttle FLY
- 16.** 200 feet RADALT Deselect IAS and reduce speed to ensure no lateral drift
- 17. If ditching:**
- Passengers **"Brace, brace, brace"**
 - Groundspeed, drift, and ROD Reduce to minimum
 - Water landing Carry out
 - Floats Deploy after touchdown
- 18. If water and aircraft conditions permit:**
- Consider extended water taxi using collective pitch for stability

Continued on next page

19. If helicopter is unstable or taking on water:

- a. Collective Lower gently
- b. Throttles STOP
- c. Rotor brake Apply gently to stop rotor
- d. Life-rafts ARM
- e. Life-raft JETT Select

Note:

The life-rafts may also be deployed by pulling the red D-rings on the forward edge of each sponson. These rings are attached to a lanyard, which must be fully extended before the raft will deploy.

20. Aircraft Evacuate; see

[14.6 EMERGENCY EVACUATION ON WATER](#)

or:

[14.7 EMERGENCY EVACUATION ON LAND](#)

----- END -----

PRE	14.6	EMERGENCY EVACUATION ON WATER
FIRE	Actions:	
ENG	1. ELT [I]	ON
APU	2. ADELT	Confirm deployed (deploy if required)
FUEL	3. Throttles	STOP
PWRTN	4. Fuel selectors	OFF
T/RTR	5. Passengers	Brief
HYD	6. Rotor brake	Apply with great caution
ELEC	7. Life-rafts	Arm / deploy when rotors stopped
CTRL	Note:	
A/ICE	<i>The life-rafts may also be deployed by pulling the red D-rings on the forward edge of each sponson. These rings are attached to a lanyard, which must be fully extended before the raft will deploy.</i>	
RIPS	8. APU CTRL	OFF
MISC	9. Battery	OFF
HOIST	10. Passengers	Evacuate into life-rafts
LAND	11. LHP	Take torch, board port life-raft
SUPP	12. RHP	Take torch, board starboard life-raft
CAS	13. Short / long painters	Cut
	14. Life-rafts	Tie together, if possible
	15. Head count	Complete
	16. Survival	Prepare
	17. PLB	Activate one per raft only
	18. If life-rafts are not deployed:	
	a. Passengers	Evacuate through starboard door
	b. Life-jackets	Connect using buddy lines
	c. PLB	Confirm transmitting
	d. Head count	Complete
	----- END -----	

- PRE
- FIRE
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- APU
- FUEL
- PWRTN
- T/RTR
- HYD
- ELEC
- CTRL
- A/ICE
- RIPS
- MISC
- HOIST
- LAND
- SUPP
- CAS

14.7

EMERGENCY EVACUATION ON LAND

- Actions:**
1. Parking brake ON
 2. ADELТ or ELТ [1] Activate
 3. PF: Radio call Complete
 4. PM: Passengers Brief
 5. Throttles STOP
 6. Rotor brake Apply
 7. APU CTRL OFF
 8. Battery OFF
 9. Cabin door Open
 - 10.Co-pilot: Passengers Evacuate upwind to safe area
 - 11.CDR: ECL / first aid kit / EFB / torches / HUMS card Collect (if possible)
 - 12.Head count Complete
 - 13.Survival Prepare
 - 14.PLB Activate one only

----- END -----

PRE
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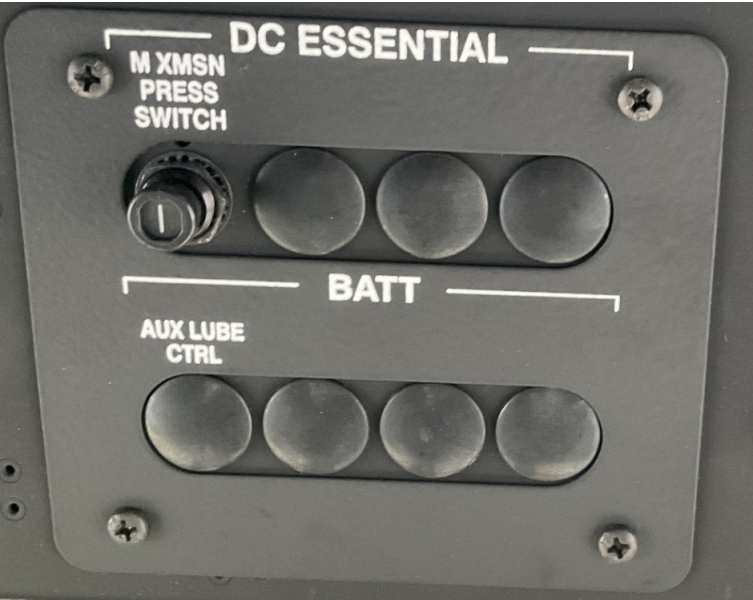
CB PANELS
CB LAYOUT
AC PRIMARY BUS 1 AND 2
DC PRIMARY BUS 1 AND 2
DC ESS AND BATT BUS
BATT HOLDUP AND BATT UTILITY BUS
ELECTRICAL DIAGRAM
WARNING AND CAUTION LIGHTS

- PRE
- FIRE
- ENG
- APU
- FUEL
- PWRTN
- T/RTR
- HYD
- ELEC
- CTRL
- A/ICE
- RIPS
- MISC
- HOIST
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SUPP. CB PANELS

1

Overhead panels:



- PRE
- FIRE
- ENG
- APU
- FUEL
- PWRTN
- T/RTR
- HYD
- ELEC
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LH panel:



- PRE
- FIRE
- ENG
- APU
- FUEL
- PWRTN
- T/RTR
- HYD
- ELEC
- CTRL
- A/ICE
- RIPS
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RH panel:



PRE

FIRE

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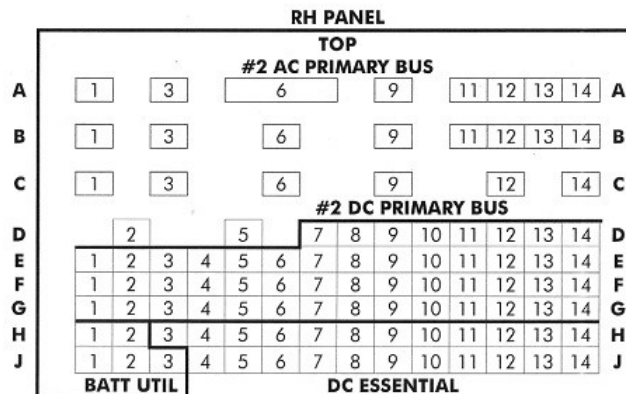
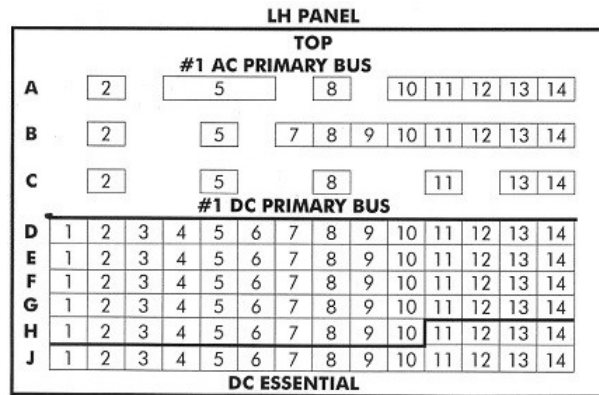
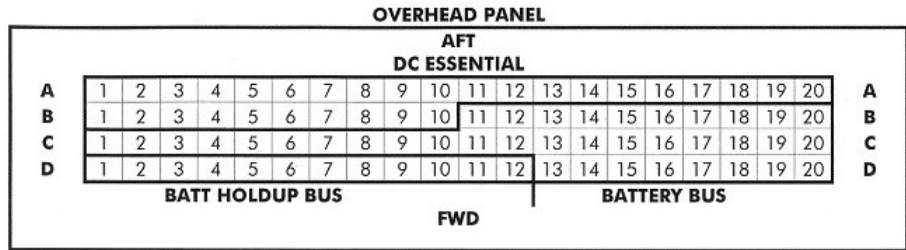
SUPP

CAS

SUPP. CB LAYOUT

2

CB PANEL LOCATION DIAGRAM



- PRE
- FIRE
- ENG
- APU
- FUEL
- PWRTN
- T/RTR
- HYD
- ELEC
- CTRL
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- CAS

SUPP. AC PRIMARY BUS 1 AND 2

3

#1 AC PRIMARY BUS	LH PANEL POS	#2 AC PRIMARY BUS	RH PANEL POS
AVC ELECS UNIT #2	B13	AVC ELECS UNIT	C3
CABIN COND BLWR	B2	CARGO HOOK LTS	A1
CABIN EVAP/VENT BLWR	C8	CKPT COND BLWR	B3
DC CONVERTER #1	A5	CKPT EVAP/VENT BLWR	A9
BU DC CONVERTER	A8	DC CONVERTER #2	A6
BU DC CONVERTER CTRL	A11	HYD FAN #2	C9
C/P WSHLD AI PWR	C2	ICE DET #2	B14
HYD FAN #1	C5	M-ADS FAN	A13
HYD GND CHK PUMP	A2	MEDEVAC	D5
ICE DET #1	B10	MFD #3 FAN	A12
MFD #1 FAN	B7	MFD #4 FAN	A11
MFD #2 FAN	B8	PLT WSHLD AI PWR	A3
MFD #5 FAN	B9	TR DEICE	C12
TR DEICE	C11	UTIL RECEPTACLE	C6
		WX RADAR	B11

SUPP. DC PRIMARY BUS 1 AND 2

4

#1 DC PRIMARY BUS	LH PANEL POS	#2 DC PRIMARY BUS	RH PANEL POS
CKPT COOL CTRL	G7	BRG MON	D13
CONV #1 SENSE	E1	CABIN UTIL RECEPT	G14
CONV #2 WARN	E2	CARGO HOOK PWR	G2
C/P ADC	D10	CARGO HOOK CTRL	G3
C/P ADF	D8	COMM #4	E14
C/P AFCS	D2	CONV #2 SENSE	D9
C/P AHRS PRI	D4	CONV #1 WARN	D8
C/P DISPLAY CTRL	E13	C/P AHRS AUX	E1
C/P DME	D6	C/P MFD INBOARD	E5
C/P FCC	D3	C/P MFD OUTBOARD	E6
C/P MAST WARN	G9	CSL BU FAN	D14
C/P MFD INBOARD	D12	CTR MFD	E3
C/P MFD OUTBOARD	D13	DOOR WARN	F1
C/P RTU	D1	ECS	F7
C/P VOR/ILS	D7	HEAT/VENT	F5
C/P WSHLD AI CTRL	F5	FDM/SSQAR	F14
C/P XPDR	D9	FIRE EXTG #2 ENG	E10
CTR MFD	D11	FMS #2	D11
DCU #1	G6	ICE DET #2	F2
USB	F1	IFE	E2
DPLR	H4	LAMP TEST	F6
EGPWS/AWG	E9	LIGHTS CHART	G13
EMER EXIT SIGNS	E14	LIGHTS HEELS	F12
FADEC 1B	G2	LIGHTS INST PNL	F8
FADEC 2B	G3	LIGHTS LWR CNSL	F9
FIRE EXTG #1 ENG	E3	LIGHTS LOGO	G10
FLOAT PWR	F7	LIGHTS #2 LTR	G11
FMS DATA LOADER	H3	LIGHTS OVHD CNSL	F10
GND CHK PUMP	G1	LIGHTS PB DIM #1	F11
HYD CTRL #3	F3	LIGHTS RTRHD	G12

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#1 DC PRIMARY BUS	LH PANEL POS	#2 DC PRIMARY BUS	RH PANEL POS
ICE DET #1	F2	LIGHTS RH CAB OHD	F13
IFE	D5	CAM PWR/M-ADS	G6
LDG LT CTRL #1	H7	MDC	D12
LDG LT PWR #1	H5	PLT RTU	F4
LDG GEAR CTRL	E5	PLT VHF	E4
LH PITOT HT PWR	G14	PLT WSHLD AI CTRL	D10
LH PITOT HT WARN	E10	RH RACK BU FAN	E9
LH RACK BU FAN	G5	RIPS CTRL #1	G5
LIGHTS CKPT RDG LTS	F10	RIPS CTRL #2	F3
LIGHTS LH CAB OVHD	F8	RSQ HOIST CTRL CBN	G1
LIGHTS LTR #1	F11	RSQ HOIST CTRL	G4
LIGHTS PASS ADV	F9	SATCOM/M-ADS	G7
LIGHTS RDG	F13	SEARCHLT CTRL	G9
LIGHTS SIDE CNSL	F12	SEARCHLT PWR	G8
LTNG DET	F14	UCD #2	E7
PARK BRAKE WARN	E4	UCD DATA LDR	E8
PASS BRIEF	H8	VHF FM	E13
PLT AHRS AUX	G10	WX RADAR CTRL	E11
PSAS #1	F6	WX RADAR R/T	E12
RAD ALT #1	E6		
RAFT PWR	G12		
RSQ HOIST CTRL BU	G13		
RSQ HOIST DIM PWR	G11		
RSQ HOIST LT CTRL	H1		
RSQ HOIST LT PWR	H2		
RIPS CTRL #1	G8		
RIPS CTRL #2	G4		
SATCOM	H6		
SATCOM CTRL	H10		
STDBY ATT	E7		

- PRE
- FIRE
- ENG
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- PWRTN
- T/RTR
- HYD
- ELEC
- CTRL
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#1 DC PRIMARY BUS	LH PANEL POS	#2 DC PRIMARY BUS	RH PANEL POS
TCAS	E8		
T/R QUAD WARN	F4		
TRIM CYC #1	D14		
UCD #1	E11		
UCD DATA LDR	E12		
WINCH CTRL	H9		

PRE

FIRE

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SUPP. DC ESS AND BATT BUS

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DC ESS BUS	L/R/O PANEL POS	BATT BUS	OVERHEAD PANEL POS
AIR/HEAT SOURCE	R J6	ACC RCG	O C7
ANTICOL LT	R J8	APU CTRL	O C4
AI BLD AIR #1 ENG	L J9	APU FIRE DET	O C11
AI BLD AIR #2 ENG	R J9	APU GEN WARN	O C2
AI WARN #1 ENG	L H13	AUX LUBE CTRL	O MGB
AI WARN #2 ENG	R J5	AVC CMPTR BU	O C9
APU BLEED VLV	O B7	BATT HOT	O D13
AVC CMPTR PRI	L J7	CARGO HOOK WARN	O C6
BAGG SMOKE DET	O B5	CONV #1 WARN	O B19
BU IND	O B9	CONV #2 WARN	O B20
CKPT COOL CTRL	O B10	C/P ICS	O D18
CLOCK	L J6	C/P RTU BU	O D16
CSL FAN	O A7	C/P VHF	O C20
DCU #2	R H8	EMER BCN	O C19
ELT	O A2	FADEC 1A	O B17
EMER LT CTRL CAB	O A11	FADEC 2A	O B18
EMER LT CTRL EXT	O A12	FDR/CVR #1	O B15
ESSL BUS SENSE	L J4	FIRE DET #1 ENG	O C17
FDR/CVR #2	R H10	FIRE DET #2 ENG	O C18
FMS #1	L H12	FIRE DET TEST	O C1
FUEL SHUTOFF #1	O A20	FIRE EXTG #1 ENG	O D15
FUEL SHUTOFF #2	L J5	FIRE EXTG #2 ENG	O D14
HYD CTR #1	R H4	FLOAT PWR	O C10
HYD CTR #2	R H3	FUEL PRIME #1	O D19
HYD WARN #1	L J13	FUEL PRIME #2	O B14
HYD WARN #2	L J14	FUEL PROBE #1	O B11
HYD WARN #3	O A17	FUEL PROBE #2	O B13
LDG LT CTRL #2	O B3	HUMS	O D17
LDG LT PWR #2	O B4	LDG GEAR EXTEND	O B12
LDG GEAR CTRL	O B1	LIGHTS BAGG COMP	O C12
LDG GEAR POS LT	O B2	LIGHTS CKPT FLOOD	O C13

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DC ESS BUS	L/R/O PANEL POS	BATT BUS	OVERHEAD PANEL POS
LH RACK FAN	O A8	LIGHTS LOAD	O C14
M XMSN PRESS SWITCH	O MGB	LIGHTS PB DIM #2	O C8
MGB OIL WARN	O A15	LIGHTS UTILITY	O C15
MGB OIL BYP	L J1	PLT ICS	O D20
OEI SEL PWR	L H14	PLT MASTER WARN	O C16
PAGE/CHIME	R H14	RAFT PWR	O C5
PLT ADC	R J10	RTR BRAKE	O C3
PLT ADF	R H12	WOW #2	O B16
PLT AFCS	R J12		
PLT AHRS PRI	R J13		
PLT DISPLAY CTRL	R H5		
PLT DME	R H13		
PLT FCC	R J11		
PLT MFD INBOARD	R H7		
PLT MFD OUTBOARD	R J7		
PLT XPDR	R H11		
PLT VOR/ILS	O A3		
POS LTS	L H11		
PSAS #2	R J14		
RAD ALT #2	R H9		
RAMP CTRL	L J8		
RAMP WARN	O B6		
RH FWD CABIN ICS	R H6		
RH PITOT HT PWR	O A18		
RH PITOT HT WARN	O A19		
RH RACK FAN	L J10		
START CTRL #1 ENG	O A9		
START CTRL #2 ENG	O A10		
STDBY PITOT PWR	L J11		
STDBY PITOT WARN	L J12		

DC ESS BUS	L/R/O PANEL POS	BATT BUS	OVERHEAD PANEL POS
TRIM COLL	O A5		
TRIM CYC #2	O A4		
TRIM YAW	O A6		
WSHLD WASH	L J2		
WSHLD WIPER	L J3		
XMSN CHIP DET	O A16		

O – OVERHEAD PANEL

O MGB – OVERHEAD MGB AUXILIARY PANEL

L – LH PANEL

R – RH PANEL

PRE

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SUPP. BATT HOLDUP AND BATT UTILITY BUS

6

BATT HOLDUP BUS	OVERHEAD PANEL POS	BATT UTILITY	RH PANEL POS
AC GEN WARN #1	D2	AC EXT PWR WARN	J1
AC GEN WARN #2	D12	ANTI COLL LT OVERRIDE	J4
APU CTRL	D6	BATT CTRL	H1
CKPT COOL CTRL	D5	FUEL PRECHECK	J3
DCU #1	D7	FUEL SELECT #1	H2
DCU #2	D8	FUEL SELECT #2	J2
MDC	D11		
PLT MFD INBOARD	D10		
PLT MFD OUTBOARD	D9		
PSAS #1	D3		
PSAS #2	D4		
WOW #1	D1		

PRE

FIRE

ENG

APU

FUEL

PWRTN

T/RTR

HYD

ELEC

CTRL

A/ICE

RIPS

MISC

HOIST

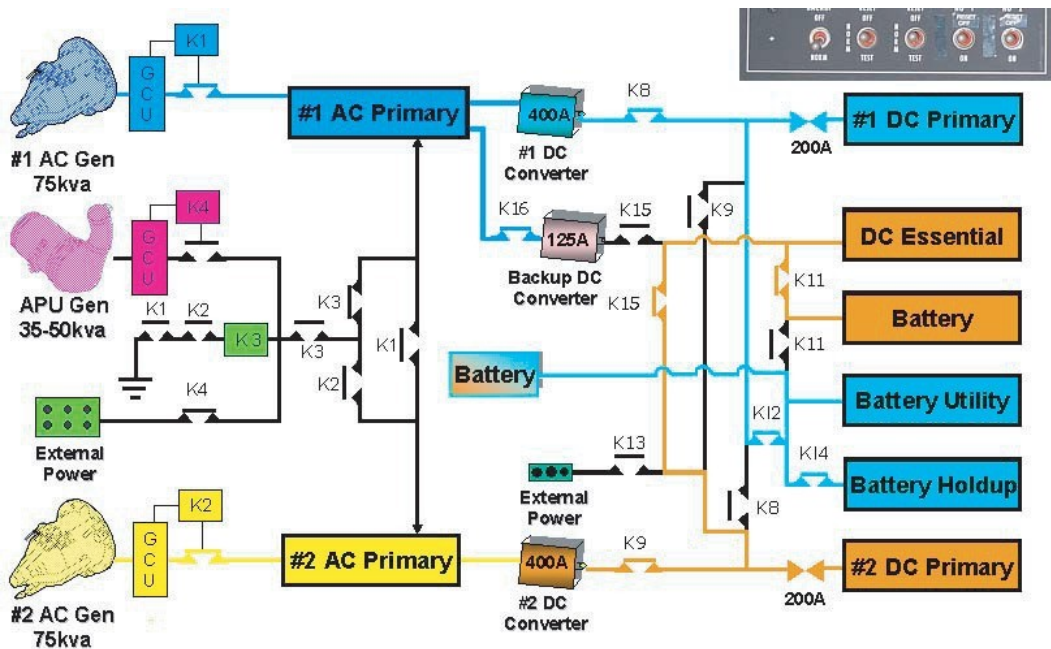
LAND

SUPP

CAS

SUPP. ELECTRICAL DIAGRAM

7



SUPP. WARNING AND CAUTION LIGHTS

8

Pilot / co-pilot master warning panels:

1 ENG OUT	FIRE	2 ENG OUT
PRESS TONE OFF	PRESS TONE OFF	PRESS TONE OFF
2.2	1.1 1.5 1.2	2.2
LO ROTOR RPM	MSTR CAUTION	LDG GEAR
PRESS TONE OFF	PRESS TO RESET	PRESS TONE OFF
2.1		7.15

Other captions and lights:

GEAR UNLKD LIGHT	CAS	CAS
7.13 7.15	12.9	12.9
DCP	DCP	DCU 1 DCU 2
12.8	12.8	12.10 12.11
	DCU 1 DCU 2	
	12.10 12.11	

EICAS caution combinations:

Cautions in the following combinations typically indicate a FAULT in the MGB caution system, rather than a failure of a mechanical component.

MGB PUMP 1 FAIL	MGB PUMP 2 FAIL
MGB OIL HOT	MGB MAN COOL
5.12	

MGB OIL PRES	MGB OIL BYPASS
5.13	

EICAS warnings and cautions are on the next page.

PRE	EICAS warnings and cautions: Attend to caution messages at bottom of EICAS page list first			
FIRE	MGB OIL PRES	5.7	FD DGRD	9.2
	MGB OIL OUT	5.8	FLOATS ARMED	12.14
ENG	SMOKE IN BAGGAGE	1.6	FUEL 1/2 BYPASS	4.2
	S/P TEMP LIMIT	5.2	FUEL 1/2 LOW	4.5
APU	ACC 1/2 CHIP	5.5	HARD LANDING	7.16
	AC GCU 1/2 FAIL	8.3	HYD 1/2 PUMP FAIL	7.1 7.2
FUEL	AC GEN 1/2 FAIL	8.1 8.2	HYD 3 PUMP FAIL	7.3 7.6
	ADS-B FAIL	12.3	HYD 1/2 RSVR LOW	7.7 7.8
PWRTN	AFCS DGRD	9.1	HYD 3 RSVR LOW	7.6
	AHRS 1/2 VEL DGRD	9.4	HYD 1/2/3 TEMP ALERT	7.5
T/RTR	APU FAIL	3.1	HYD 1/2/3 TEMP LIMIT	7.5
	APU FAULT	3.2	HYDR FAN FAIL	7.4
HYD	APU GCU FAIL	3.6	ICE DETECTED	11.7
	APU GEN FAIL	3.5	IGB CHIP	5.4
ELEC	APU OVERLOAD	3.4	IGB HOT	5.4
	AUTO DEICE FAIL	11.13	IGB OIL LEVEL	5.3
CTRL	AUTOPILOT 1/2 FAIL	9.8	INLT 1/2 A-ICE FAIL	10.6
	AUX XFER DGRD	4.8	INPUT 1/2 CHIP	5.5
A/ICE	AUX XFER FAIL	4.7	INPUT / ACC 1/2 HOT	5.9
	AVC DEGRADED	12.4	LIMIT EXCEEDED	2.19
RIPS	AVC FAIL	12.4	MAINT COMP FAIL	12.17
	BACKUP CONV OFF	8.10	MGB OIL BYPASS	5.16
MISC	BATT CHARGE FAIL	8.5	MGB CHIP	5.5
	BATT HOT	8.4	MGB MAN COOL	5.15
HOIST	BATT OFF	8.11	MGB OIL HOT	5.11
	BOOST PRES	7.11	MGB OIL PRES	5.6
LAND	BYPASS TEST FAIL	5.14	MGB PUMP 1/2 FAIL	5.10
			MGB SENSOR FAIL	5.17
SUPP	CHECK RIPS MODE	11.10	MR DEICE FAIL	11.11
	CHIP SYS FAULT	12.5	MR DEICE FAULT	11.8
CAS	COCKPIT SEC FAN	12.7	MR SVO 1/2 PRES	7.9
	CONFIG MISCOMP	12.6	PITOT 1/2/3 HEAT	10.1
	DC CONV 1/2 FAIL	8.7 8.8 8.9		
	DCU MISCOMPARE	12.10	PRIME PUMP FAIL	4.3
	DEICE CNTLR FAIL	11.4	PSAS 1/2 FAULT	9.7
	DEICE SYS FAIL	11.1	RACK 1/2 FAN FAIL	12.7

PRE	DOOR OPEN	12.12	RAFTS ARMED	12.19
	DROOP HTR FAIL	11.2	RAMP OPEN	12.20
FIRE	ENG 1/2 A-ICE DGRD	10.5	RIPS DEGRD LWC	11.6
	ENG 1/2 CHIP	2.6	RIPS DEGRD OAT	11.9
ENG	ENG 1/2 FIXED PWR	2.18	ROTOR BRAKE	12.21
	ENG 1/2 FUEL PRES	4.1	SAS 1/2 PRES	7.12
APU	ENG 1/2 HOT START	2.4	SOV 1/2 WRNG POSN	4.4
	ENG 1/2 OIL BYPASS	2.5	S/P TEMP DETECT	5.1
FUEL	ENG 1/2 OIL HOT	2.5	STARTER 1/2 ON	2.21
	ENG 1/2 OIL PRES	2.5	STOP AUX XFER	4.6
PWRTN	ENG 1/2 OVERSPEED	2.3	TGB CHIP	5.4
	ENG 1/2 OVSP TEST	2.20	TGB HOT	5.4
T/RTR	ENG 1/2 POWER LIM	2.16	TGB OIL LEVEL	5.3
	ENG 1/2 RATE LIM	2.17	TR QUAD	6.2
HYD	FADEC 1/2 A/B FAIL	2.13 2.14	TR SVO 1/2 PRES	6.3 7.10
			TRAIN DISABLED	2.22
ELEC	FADEC 1/2 NO DSPTCH	2.11	TRN PIT HEAT ON	10.2
	FADEC 1/2 FAULT	2.10	TRN PIT HEAT OFF	10.3
CTRL	FADEC 1/2 OVRHEAT	2.12	TR DEICE FAIL	11.12
			TRN INLT A-I ON	10.4
A/ICE	FADEC 1/2 XENG	2.15	TURN RIPS ON	11.3
	FD 1/2 ALT DGRD	9.3		
RIPS				
MISC				
HOIST				
LAND				
SUPP				
CAS				